Table of Contents

AEON: An Enterprise Control Plane Architecture for the Agentic Era

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Abstract

This paper specifies AEON (AI Enterprise Orchestration Nexus) as the enterprise control plane architecture for the agentic era. AEON is the architectural layer that orchestrates capability composition across the enterprise's subdomains, owns the identity and authority fabric that flows across them, captures evidence at enterprise scope, decomposes policy from enterprise scope into subdomain-specific enforcement, manages the capability lifecycle, and provides the meta-orchestration runtime that coordinates the enterprise's operational picture as an integrated system rather than as a collection of silos. The architectural posture is that AEON is Enterprise IT-developed and sustained institutional infrastructure, not a vendor product the enterprise integrates. Components are vendor-supplied where appropriate; the composition of those components into the AEON architecture is Enterprise IT's responsibility. The strategic claim is that the agentic era requires an enterprise control plane, that the control plane must be enterprise-owned to preserve institutional governance over enterprise operations, and that Enterprise IT is the natural owner of this work. The paper reads at two altitudes. An executive layer presents the value case, the institutional preconditions an enterprise must meet before committing, and a worked scenario that makes the architecture legible to a sponsor. An implementer layer specifies the six service planes (identity, authority, evidence, integration, capability composition, orchestration runtime), defines the Enterprise IT operating model required to build and sustain AEON, addresses the multi-classification environments defense work operates in, treats the relationship between AEON and existing enterprise investments, sketches a phased deployment path that includes a minimal coherent subset delivering value on its own, and surfaces the open questions, concerns, and opportunities the architecture creates with concrete mechanisms for the governance and change-management risks. The paper is addressed primarily to enterprise leadership at the CIO and Chief Architect level, with implications for CISO, identity engineering, enterprise architecture, and the subdomain engineering functions that compose into AEON.

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1. The Enterprise Control Plane Problem

The enterprise architecture that the defense industrial base operates today was assembled over three decades. It is organized around systems of record at the center (ERP, PLM, CRM, engineering data management), enterprise integration platforms that connect them, identity and access management infrastructure that authorizes access to them, and a layer of department-level applications and platforms that sit on top. This architecture absorbed two prior technology waves reasonably well. The web wave produced enterprise portals that overlaid existing systems without disturbing them. The cloud wave produced lift-and-shift migrations that preserved the architectural pattern while changing the underlying infrastructure. In both cases, the architectural decomposition survived because neither wave fundamentally changed what enterprise composition meant.

The agentic wave is different. The unit of enterprise composition is no longer the system of record accessed by a human user. It is the agentic capability invoked by a human principal whose authority chain has to be preserved through the agent, through whatever tools the agent invokes, through whatever other agents those tools call, and back to the originating principal in a verifiable way. The composition pattern that worked for human-driven access to systems of record does not absorb agentic execution. It cannot, because it was not designed to.

The naïve response is to layer agentic capability onto existing enterprise architecture as if it were just another category of application. This response is being attempted across the industry, and it is producing visible failures. The failures are not technical in the narrow sense; the individual agentic products work. The failures are architectural. Identity does not flow correctly across agents. Authorization decisions cannot be made at the right altitude because the policy infrastructure was not built to evaluate agentic action. Evidence does not aggregate into an enterprise audit picture because the systems generating evidence were not designed to feed an enterprise evidence plane. Cross-subdomain workflows that the agentic substrate makes possible cannot be governed because the integration infrastructure was not built to orchestrate them. The agentic capabilities exist, but the enterprise has lost the ability to govern their composition.

The architectural response is a control plane. A control plane is the architectural layer that orchestrates composition without executing the composed capabilities. The data plane runs the actual workloads; the control plane decides what is composed, by whom, under what authority, with what evidence, against what policy. The control plane pattern is well established in networking, in cloud infrastructure, in container orchestration, in microservices architecture. It is not established in enterprise architecture for the simple reason that enterprise architecture has not previously needed one. The pre-agentic enterprise composed capabilities through human users invoking applications, with the human serving as the control plane in effect. The agentic enterprise cannot rely on this pattern because the human is no longer present at every composition decision; agents are composing other agents, tools, and systems on the human's delegated authority, faster and at greater scale than the human can supervise.

AEON is the enterprise control plane for the agentic era. The name encodes the architectural posture. AI is what the control plane is for; the agentic era is what makes the control plane necessary. Enterprise is the scope; the control plane operates at enterprise altitude, not at department altitude or program altitude. Orchestration is the function; AEON orchestrates the composition of capabilities, not the execution of any individual capability. Nexus is the architectural role; AEON is the point where the enterprise's composition decisions converge. The control plane is not the platform on which the enterprise runs; it is the architectural layer that makes the enterprise composable in the first place.

This paper specifies AEON in enough detail to be built. The specification is at the architectural altitude appropriate to a control plane: not down to the protocol or the wire format, but down to the planes, the service definitions, the composition patterns, and the operating model. Enterprise IT functions that develop and sustain AEON will translate the architecture into specific engineering decisions, vendor selections, and integration designs appropriate to their context. The architecture is portable across those decisions; the engineering is not.

The paper is positioned as an architectural argument rather than a vendor-engaged or literature-engaged scholarly treatment. The control plane pattern as applied to enterprise agentic AI is novel enough that the literature is still forming; the vendor landscape is still consolidating; the operating practice is still being invented across the industry. The paper's contribution is the architectural specification of AEON as the enterprise's coherent response to the agentic composition problem, developed in enough detail that an enterprise can commit to building it.

The paper reads at two altitudes. Sections 2 through 4 are the executive layer: the value case, the institutional preconditions, and a worked scenario that make the architecture legible to a sponsor in a single sitting. Sections 5 through 10 are the implementer layer: the plane specifications and the operating model that an architecture team builds against. Sections 11 through 16 address deployment, classification, existing investments, phasing, and the open questions. An executive can read Sections 2 through 4 and the conclusion and understand what is being proposed and what it requires; an implementer reads the whole.

2. The Executive Case

This section is for the sponsor who has to decide whether the enterprise commits to AEON. It states the value, the alternative, and the economics in terms that do not require the plane model that follows.

2.1 The Decision

Every enterprise is acquiring agentic AI capability whether or not it has decided to. Productivity substrates, coding agents, analytics copilots, and workflow automations are entering the enterprise through vendor products, individual programs, and departmental initiatives. The capability is arriving. The only open question is whether the enterprise governs its composition deliberately or lets the composition emerge from whatever the vendors and the programs happen to produce.

The decision in front of the sponsor is therefore not whether to adopt agentic AI. That decision is being made continuously by the enterprise's own momentum. The decision is whether to build an enterprise control plane that governs the composition, or to accept the composition that emerges by default. AEON is the deliberate version of the choice. The default version is what happens when no one makes the choice.

2.2 The Two Futures

Consider two enterprises three years from now. Both adopted agentic AI aggressively. One built an enterprise-owned control plane; one did not.

The enterprise without a control plane has working agentic products throughout its operations. It also has identity that does not flow correctly across the products, so it cannot reliably say which human authorized which agentic action. It has authorization decisions made inconsistently across products, because each product enforces its own policy in its own way. It has evidence scattered across product-specific logs that do not correlate, so investigating an agentic incident requires manual reconciliation across systems that were never designed to be reconciled. It has cross-product workflows that no one can govern because no architectural layer composes them. When a regulator asks the enterprise to demonstrate governance over its agentic operations, the enterprise cannot, because the governance was never architected. The enterprise responds with procedural controls, compliance overhead, and policy documents that add friction without restoring the architectural integrity that was never built.

The enterprise with a control plane has the same working agentic products. It also has identity that flows across all of them through a single federation, so every agentic action traces to a human principal. It has authorization evaluated consistently at enterprise altitude. It has evidence that aggregates into a single audit picture queryable across all products. It has cross-product workflows that compose under governance. When a regulator asks the enterprise to demonstrate governance over its agentic operations, the enterprise demonstrates it through the architecture. The governance is an architectural property rather than a procedural overlay.

The difference between the two futures is not the agentic capability. Both have it. The difference is whether the enterprise can govern the capability it has. The control plane is what makes the capability governable.

2.3 The Business Outcomes

The architecture produces specific business outcomes a sponsor can hold it accountable for.

Audit time-to-investigate after an agentic incident. In the enterprise without a control plane, investigating what an agentic workflow did across multiple subdomains is a manual reconciliation across uncorrelated logs, measured in days or weeks, often inconclusive. In the enterprise with AEON, the same investigation is a query against a correlated evidence picture, measured in hours, with a definitive answer. The reduction in time-to-investigate and the increase in conclusiveness are both direct consequences of the evidence plane.

Certifiable compliance under agentic operation. Defense compliance regimes (CMMC, ITAR, EAR, program-specific requirements) increasingly have to be demonstrated for agentic operations, not just for human-driven operations. Without a control plane, the enterprise cannot demonstrate that agentic actions were authorized, traced, and governed, because the architectural mechanisms for the demonstration do not exist. With AEON, compliance under agentic operation is demonstrable through the identity, authority, and evidence planes. The difference is the ability to certify, not merely to assert.

Capability onboarding time under governance. Introducing a new agentic capability into the enterprise without a control plane is fast but ungoverned; the capability works but the enterprise cannot say how it composes with everything else. Introducing it with AEON requires the capability to register, qualify, and expose a compliant adapter, which is slower initially but produces a governed capability that composes coherently. The control plane trades some onboarding speed for onboarding governance; the trade is favorable for any capability consequential enough to require governance, and the architecture's risk-tiering allows low-consequence capabilities to onboard with lighter governance.

Vendor optionality preserved. Without a control plane, the enterprise's agentic capability is distributed across vendor products that each own a piece of the composition, producing incremental lock-in that compounds over time. With AEON, vendors supply components within an enterprise-owned architecture, and the enterprise retains the ability to replace any component without rebuilding its composition. The preserved optionality has option value that is real even when it is not exercised.

2.4 The Economics

A sponsor will ask what AEON costs against what buying a vendor agentic platform costs. The honest answer is that the comparison cannot be reduced to a single defensible number for a generic enterprise, because both sides of the comparison depend heavily on enterprise-specific factors: the maturity of existing investments, the scale of agentic adoption, the regulatory exposure, and the cost of the senior talent the function requires. What can be specified is the structure of the comparison, which the enterprise then populates with its own figures.

On the cost side, AEON requires the standing function (a small senior team, the most expensive line item, sustained over a multi-year horizon), the runtime engineering (substantial but bounded, composed largely of vendor components), the policy translation program (variable, depending on how much enterprise policy has to move into policy-as-code), and the integration effort across subdomains (paced and incremental). The vendor-platform alternative requires licensing (recurring, scaling with usage), integration to the vendor platform (substantial and vendor-specific), and the accepted lock-in (a cost that does not appear on the invoice but is real).

On the value side, AEON produces the governance outcomes in Section 2.3, the preserved optionality, and the institutional capability that compounds over time. The vendor platform produces faster initial capability and lower initial cost, offset by the governance gaps, the lock-in, and the eventual cost of recovering architectural control if the vendor relationship sours or the vendor's trajectory diverges from the enterprise's needs.

The comparison axes that matter are time-to-initial-capability (vendor platform wins), time-to-governed-capability (AEON wins), total cost over a multi-year horizon (depends heavily on scale and lock-in cost), strategic optionality (AEON wins decisively), and governance certifiability (AEON wins decisively). The enterprise has to weight these axes according to its own situation. An enterprise with low regulatory exposure and modest agentic ambitions may reasonably weight time-to-initial-capability heavily and choose the vendor platform. A defense enterprise with high regulatory exposure and substantial agentic ambitions should weight governance certifiability and strategic optionality heavily, which favors AEON.

The economic argument is therefore not that AEON is cheaper. It is that AEON is the correct investment for an enterprise whose regulatory exposure and strategic posture make governance certifiability and architectural optionality more valuable than time-to-initial-capability and minimized initial cost. The defense industrial base is squarely in that category. Enterprises outside it should run the comparison honestly and may reach a different conclusion.

3. Institutional Preconditions

AEON is not appropriate for every enterprise. The architecture presumes institutional conditions that some enterprises have, some can develop, and some neither have nor will develop. A sponsor should evaluate the enterprise against these preconditions before committing. An enterprise that lacks them and will not develop them should not attempt AEON; partial commitment produces a partial control plane, which is worse than a deliberate decision to govern agentic composition some other way.

The preconditions are the following.

A consolidated enterprise IDAM. The identity plane federates across IDAM instances, but the federation broker can mediate a few authorities, not dozens. An enterprise whose identity infrastructure is fragmented across many uncoordinated authorities has to consolidate before AEON is tractable. The precondition is enterprise IDAM with a small number of primary authorities per classification environment.

Willingness to invest in policy-as-code. The authority plane evaluates policy as code at runtime. An enterprise whose policy lives entirely in documents and human governance processes has to build policy-as-code capacity, which requires legal, compliance, and IT to author policy jointly in a form that did not previously exist in the enterprise. The precondition is institutional willingness to make this investment, not its prior existence.

A central architecture function with real authority. AEON's contracts bind subdomains. An enterprise whose architecture function is purely advisory, with no authority over how subdomains compose, cannot enforce the contracts the architecture requires. The precondition is a central architecture function with genuine authority over cross-domain contracts, backed by senior leadership.

Tolerance for a multi-year horizon. AEON is a multi-year program before it reaches enterprise coverage. An enterprise whose investment discipline cannot sustain a multi-year infrastructure program through budget cycles and leadership transitions will abandon AEON partway, which is worse than not starting. The precondition is institutional tolerance for a three-to-five-year horizon before full coverage.

Ability to recruit and retain senior control-plane talent. The function requires senior architects and engineers with skills that are scarce and expensive. An enterprise that cannot recruit or retain this talent, whether because of compensation constraints, location constraints, or clearance constraints, cannot staff the function. The precondition is the ability to assemble and sustain the team.

These preconditions are an honest filter. An enterprise that has them, or has a credible plan to develop them, is a candidate for AEON. An enterprise that lacks them and has no plan to develop them should govern its agentic composition through a different and less ambitious approach, accepting the governance limitations that approach entails, rather than committing to an architecture it cannot sustain.

4. A Worked Scenario

Before the plane specifications, a single worked scenario shows the planes operating together. The scenario is deliberately ordinary; the architecture's value is most visible in routine work, not in exotic cases.

A systems architect, authenticated through enterprise IDAM, asks her AIDEX substrate to assess a system architecture against current security guidance and propose remediations. This single request composes across four subdomains. The annotated sequence shows which plane does what.

Step 1, Identity plane. The architect authenticated at session start; AEON resolved her to a principal carrying her clearance, program assignments, and role. The authority token rooted in that authentication is the basis for everything that follows. Her AIDEX substrate acts under her delegated authority, carrying her identity in every downstream action.

Step 2, Authority plane. The substrate proposes to retrieve the system architecture from Digital Engineering. AEON's authority plane evaluates the proposal against the architect's current attributes and the policy governing the architecture's classification. The decision returns allow, because her clearance and program assignment authorize access to this architecture.

Step 3, Integration plane. The substrate's request to Digital Engineering crosses a subdomain boundary. The integration plane composes the cross-subdomain call, extending the delegation chain so Digital Engineering receives the architect's identity and verifies her authority before returning the architecture. The chain integrity is preserved across the boundary.

Step 4, Authority plane, compose-and-evaluate. The substrate next proposes to consult current security guidance from Cybersecurity and applicable compliance requirements from Governance and Strategy, then synthesize a remediation proposal. This composes authorities across four subdomains. No single subdomain can authorize the composite action; the authority plane composes the relevant policy across all four, evaluates the composition, and returns a decision that all four can rely on.

Step 5, Integration plane, workflow composition. The integration plane orchestrates the composite workflow: retrieve the architecture, consult the security guidance, check the compliance requirements, and return the synthesized inputs to the substrate. The workflow is composed as a single governed unit, not as four independent actions.

Step 6, Evidence plane. Every step contributes canonical evidence: the action attempted, the principal who authorized it, the authority decision that sanctioned it, the subdomain that performed it, the artifacts touched, and the timestamp chain. The evidence aggregates into the enterprise audit picture, correlated through the architect's delegation chain.

Step 7, AIDEX, propose-and-confirm. The substrate presents the remediation proposal to the architect. She curates it: she accepts some remediations, rejects others, modifies the rest. Her curation is captured cryptographically and returned to the evidence plane as part of the record. The remediation she signs is her curated work, not the substrate's unreviewed output.

The whole sequence took seconds. The architect experienced it as asking a question and getting a well-formed proposal she could curate. Underneath, four subdomains composed under enterprise governance, with identity flowing throughout, authority evaluated at the right altitude, the workflow composed as a governed unit, and evidence captured for the enterprise audit picture. If the same sequence happened in an enterprise without a control plane, the architect might get a similar proposal, but the enterprise would have no architectural record of which human authorized the cross-subdomain access, no consistent authorization evaluation, and no correlated evidence. The proposal would arrive; the governance would not.

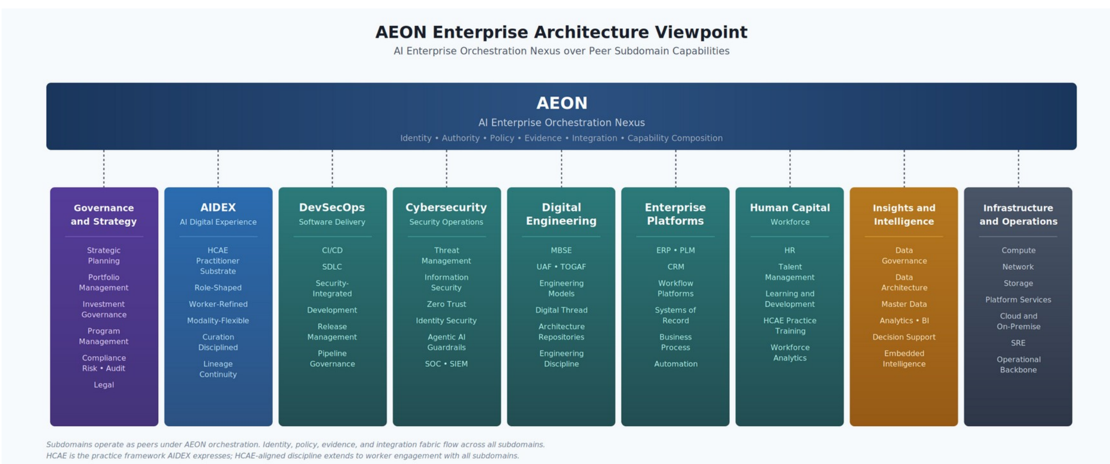
The plane specifications that follow develop each plane the scenario exercised. The scenario is the reference the specifications return to.

5. Locating AEON in the Enterprise Architecture

AEON is the control plane. The subdomains are the capability areas the control plane orchestrates. The relationship is hierarchical in the architectural sense but not in the operational sense: AEON does not run the subdomains, and the subdomains do not report to AEON's runtime. The relationship is that AEON specifies how the subdomains compose, and the subdomains expose capabilities that comply with AEON's composition contract.

The canonical subdomain set is nine peer subdomains: Governance and Strategy, AIDEX (the worker-facing AI digital experience substrate developed in the companion paper), DevSecOps, Cybersecurity, Digital Engineering, Enterprise Platforms, Human Capital, Insights and Intelligence, and Infrastructure and Operations. Each is a capability area the control plane orchestrates; none is privileged over the others. The full scope of each subdomain is described in Appendix A; the viewpoint below shows the set under AEON orchestration.

The viewpoint of this architecture is presented below.



AEON Enterprise Architecture Viewpoint

The viewpoint expresses several architectural commitments that the rest of the paper develops. The subdomains operate as peers under AEON orchestration. None of them is privileged in the architecture; each contributes capabilities that AEON composes. The identity, policy, evidence, and integration fabric flows across all subdomains; it is not a property of any one of them. AEON sits above the subdomains because it orchestrates them, but the architectural altitude is meta-orchestration, not super-platform. AEON does not run the subdomains; it specifies how the subdomains compose into the enterprise's operational picture.

What AEON does. AEON owns the identity plane: the IDAM federation across classification environments, the principal identity resolution that the subdomains consume, the delegation chain integrity that flows through every action across every subdomain. AEON owns the authority plane: the policy decomposition from enterprise scope into subdomain-specific enforcement, the authorization decision protocol the subdomains consult, the cross-subdomain authorization composition that no single subdomain can perform on its own. AEON owns the evidence plane: the aggregation of evidence from subdomain activity into the enterprise audit picture, the integrity protections that make the evidence forensically meaningful, the worker-facing evidence access that protects principal accountability. AEON owns the integration plane: the cross-subdomain workflow orchestration, the agent network composition across subdomains, the delegation chain integrity across subdomain boundaries. AEON owns capability composition: the capability registry, the composition contract, the lifecycle management for capabilities the enterprise exposes through any subdomain. AEON owns the meta-orchestration runtime: the actual platform that Enterprise IT builds and operates, the registry of subdomain capabilities, the policy engine integration, the evidence aggregation runtime, and the operational backbone of the control plane itself.

What AEON does not do. AEON does not run DevSecOps pipelines. AEON does not perform threat detection. AEON does not maintain engineering models. AEON does not process payroll. AEON does not host application workloads. AEON does not produce decision support analyses. AEON does not perform any of the subdomain functions. Each subdomain has its own runtime, its own internal architecture, its own engineering function, and its own vendor relationships. The subdomain teams are not part of the AEON team. The AEON team builds and operates the control plane; the subdomain teams build and operate their subdomains and compose into AEON through the architectural contracts AEON specifies.

This separation matters. The temptation to make AEON do everything must be resisted, for two reasons. The architectural reason is that AEON's value as a control plane depends on staying at the meta-orchestration altitude; once AEON starts running subdomain workloads, it becomes another platform competing with the subdomains rather than the architectural layer that organizes them. The operational reason is that the AEON team cannot be staffed or scoped to absorb subdomain operations; the team has to be small, senior, architecturally focused, and authoritative within its scope. A control plane that tries to do everything cannot do anything well.

The rest of the paper develops AEON at the meta-orchestration altitude. Subdomain-internal architecture is referenced where the relationship to AEON requires it, but the subject of the paper is the control plane, not the subdomains.

6. The Architectural Posture

AEON is Enterprise IT-developed and sustained institutional infrastructure. This is the architectural posture, and it shapes every subsequent specification decision. The posture has three components that need to be made explicit because they are not the default assumption in many enterprises and because they have substantial implications for how AEON is built, operated, and evolved.

6.1 Institutional Ownership

AEON is owned by the enterprise. The control plane that orchestrates the enterprise's agentic capability is too consequential to outsource. Identity, authority, policy, evidence, and capability composition are the architectural mechanisms by which the enterprise exercises governance over its own operations. A vendor-supplied control plane means vendor governance over enterprise operations, mediated through whatever contractual and product decisions the vendor makes. For a defense industrial enterprise operating under explicit regulatory accountability for its operations, vendor governance over the control plane is unacceptable. For any large institution that takes its own accountability seriously, vendor governance over the control plane should also be unacceptable, though the recognition of this is often delayed until the consequences become visible.

The control plane is institutional infrastructure in the same sense that the enterprise integration architecture, the enterprise identity architecture, and the enterprise data architecture are institutional infrastructure. These are functions the enterprise builds and sustains because they encode enterprise-specific governance commitments that no vendor can encode on the enterprise's behalf. The enterprise does not outsource its own architecture. It outsources components within its architecture, which is a different thing.

Vendor lock at the control plane is also catastrophic in a way that vendor lock at the subdomain level is not. The enterprise can replace a CRM with another CRM. It can replace a project management tool with another. It can replace an AI productivity substrate with another (the substrate portability commitment in the AIDEX architecture depends on this option remaining open). What the enterprise cannot easily replace is the control plane that all of those plug into. If the control plane is vendor-owned, every subdomain decision becomes contingent on the vendor's continued cooperation, pricing, capability investment, and strategic posture. The enterprise has surrendered architectural control over its own composition. Recovering that control after the fact requires rebuilding the control plane from scratch with all subdomains rebound to it, which is a multi-year program at enterprise scale.

The strategic argument is therefore that the control plane has to be enterprise-owned, and the operational argument is that Enterprise IT is the natural owner. Enterprise IT is the function within the enterprise whose scope is enterprise infrastructure, whose mandate is institutional capability rather than departmental capability, and whose accountability is to the enterprise rather than to any particular business unit. The control plane is enterprise infrastructure by definition; Enterprise IT is the enterprise's infrastructure owner; therefore AEON is Enterprise IT's responsibility.

6.2 Compositional Construction

Enterprise IT-built does not mean from scratch. The “developed and sustained” framing means Enterprise IT owns the architecture, owns the composition, owns the lifecycle, owns the operations. It does not mean Enterprise IT writes every line of code that runs in the AEON runtime. Components within the AEON architecture can be, and in most cases should be, vendor-supplied. Identity infrastructure from established identity providers. Policy engines from open-source policy projects or commercial policy platforms. Evidence stores from commercial SIEM and audit platforms or open-source equivalents. Integration runtimes from established integration platforms. Capability registries built internally because the registry encodes enterprise-specific composition decisions that vendors cannot encode generically.

The composition of these components into the AEON architecture is Enterprise IT’s responsibility. The architectural decisions about which planes exist, what they expose, how subdomains consume them, and how they compose into the enterprise operational picture are not vendor decisions. The engineering decisions about how to translate the architecture into running infrastructure are Enterprise IT decisions, informed by vendor capabilities but not dictated by them. The lifecycle of AEON, including how it evolves, how it scales, how it absorbs new capabilities and retires old ones, is Enterprise IT’s responsibility.

This compositional posture has a name in enterprise architecture practice. It is the same posture mature enterprises take toward their integration architecture: the enterprise owns the architecture, the integration platforms are vendor-supplied, the integrations themselves are designed and operated by Enterprise IT. The result is institutional infrastructure that is portable across vendor changes, evolves under enterprise control, and embodies enterprise-specific architectural commitments that survive vendor decisions. AEON follows this same pattern at the control plane altitude.

The compositional posture also implies vendor-component-friendliness coupled with vendor-control-plane-incompatibility. The architecture welcomes vendor components within the planes it defines; the architecture rejects vendor control planes that would replace the AEON architecture itself. This is not a contradiction. Vendors offering identity providers, policy engines, evidence stores, integration runtimes, and capability platforms are providing components Enterprise IT composes into AEON. Vendors offering “AI orchestration platforms” or “agentic enterprise platforms” that claim to be the control plane are offering an architectural substitute for AEON, and the substitute is incompatible with enterprise ownership. The distinction is whether the vendor is providing a component or claiming the architecture; the former is supported, the latter is rejected.

6.3 Capability Discipline

The Enterprise IT function that develops and sustains AEON is not the same as the Enterprise IT function that ran the pre-agentic enterprise. The capabilities required for control plane work differ in kind from the capabilities required for traditional application delivery, infrastructure operation, or platform engineering. Identity engineering at the level AEON requires is a different discipline from access management as practiced in most enterprises today. Policy as code at enterprise altitude is a different discipline from policy authoring in traditional governance functions. Evidence engineering for agentic action is a different discipline from log management and audit support. Capability composition as an architectural discipline is a different discipline from enterprise architecture as practiced in most TOGAF-aligned functions today.

The paper does not pretend the Enterprise IT function is ready for this work as constituted today. The capability uplift required is substantial. Section 13 develops the operating model in detail; what the strategic posture asserts here is that the enterprise is committing to develop this function, not pretending it already exists. Enterprises that are not willing to develop this function should not commit to building AEON, because the resulting platform will be unsound and the enterprise will be worse off than if it had pursued a different architectural approach.

This is also why the architectural posture has to be made explicit early. The decision to commit to AEON is the decision to commit to the institutional ownership, the compositional construction, and the capability discipline. The architecture cannot be successfully adopted without all three. Enterprises that try to adopt the architecture without committing to the function will produce a platform that fails operationally, fails to provide the governance the architecture promises, and damages the enterprise's posture in the agentic era more than non-adoption would have.

6.4 The Strategic Position

The enterprise that develops AEON well, with the institutional ownership, the compositional construction, and the capability discipline, achieves a strategic position the agentic era will reward. The enterprise's agentic capability composes coherently. Identity flows across subdomains. Authority is evaluated consistently. Evidence aggregates into an enterprise audit picture. Cross-subdomain workflows are governable. Capability composition is a deliberate institutional act rather than an accidental emergent property. The enterprise can absorb new capabilities under coherent governance, and can retire old ones without leaving accountability gaps.

The enterprise that does not develop AEON, or develops it badly, finds the opposite. Agentic capability accumulates without coherent composition. Identity does not flow correctly across subdomains. Authority cannot be evaluated at the right altitude. Evidence does not aggregate. Cross-subdomain workflows are ungovernable. Capability composition is whatever vendors and individual programs happen to produce. The enterprise's governance posture erodes as agentic capability expands. The conventional response is more policy, more compliance overhead, more procedural controls, all of which add friction without restoring the architectural integrity that was never there to begin with.

The strategic argument is therefore not that AEON is one option among several. The argument is that the agentic era requires an enterprise control plane regardless of any individual enterprise's choice, and the choice is whether the control plane is deliberately architected and owned, or accidentally emerges from vendor decisions made without enterprise architectural intent. AEON is the architectural framework for the deliberate version of this choice. The rest of the paper specifies what the deliberate version actually entails.

7. The Identity Plane

The identity plane is where AEON begins, because every other plane depends on it. Authority is what an identified principal is permitted to do. Evidence is what an identified principal did. Integration is what one identified principal's agent does on behalf of another identified principal. Capability composition is what identified principals can combine into workflows. Without identity, none of these planes have meaning. AEON's identity plane is the foundation that makes the rest of the architecture coherent.

7.1 What the Identity Plane Owns

The identity plane owns the resolution of any subdomain action to the principal that authorized it. The principal is a human person identified through enterprise IDAM infrastructure, typically via CAC, PIV, or equivalent credentials processed through a federated identity provider that returns attribute claims about the person. The attribute claims include clearance level, program assignments, role designations, organizational position, and access scopes appropriate to the enterprise's operations.

The identity plane is the source of truth for principal identity across all subdomains. AIDEX, DevSecOps, Cybersecurity, Digital Engineering, Enterprise Platforms, Human Capital, Insights and Intelligence, and Infrastructure and Operations all consume identity from the AEON identity plane. They do not maintain their own user stores. They do not authenticate principals themselves. They accept the AEON-resolved principal as the authoritative identity for any action they perform on the principal's behalf.

This is a strong architectural commitment, and it is the central commitment of the identity plane. Subdomains that maintain their own user stores produce identity fragmentation that makes cross-subdomain governance impossible. The same person becomes different principals in different subdomains, with no architectural mechanism for asserting that they are the same person. The identity plane's job is to make this fragmentation impossible by being the authoritative source.

7.2 The Federation Topology

Enterprise IDAM is rarely a single infrastructure. Defense enterprises typically operate multiple IDAM instances across classification environments: an unclassified IDAM, one or more CUI IDAMs, separate IDAMs for NOFORN or restricted programs, and classified IDAMs that may themselves be federated across program boundaries. The identity plane has to operate across this federated topology without producing identity fragmentation across the boundaries.

The architectural pattern AEON uses is a federation broker. The broker is not an IDAM itself; it is an architectural service that mediates between IDAMs while preserving each IDAM's authority over its own principals. A principal authenticated in an unclassified IDAM has a derived identity recognizable in a CUI environment. A principal authenticated in a CUI IDAM has a derived identity recognizable in an unclassified environment, possibly with attribute redaction appropriate to the classification differential. A principal who exists in a NOFORN IDAM has no derived identity in unclassified environments, because the NOFORN status itself is information that does not propagate down.

The federation patterns are several, and the deployment selects from them based on operational requirements. *Derived identities* establish a relationship between principal identities in different domains such that a principal authenticated in one domain has a verifiable corresponding identity in another, without requiring synchronous authentication across the boundary. *One-way attribute synchronization* propagates principal state from a higher-classification domain to lower-classification domains, allowing the lower domain to recognize the higher-domain principal without granting the higher domain dependency on the lower. *Portable capability tokens* issued by a higher-trust IDAM and carrying constrained scope can be presented to lower-trust environments for specific authorized operations, with cryptographic constraints preventing scope expansion. *Cross-domain identity brokers* mediate between IDAM instances with explicit policy controls on what propagates and in what direction.

The AEON identity plane integrates these patterns rather than selecting one. Different federation relationships use different patterns based on what the operational and security posture requires. The architectural commitment is that the federation broker exists, is governed under AEON's control plane, and preserves the accountability chain across every boundary it crosses. Implementations vary by enterprise; the architecture is portable across implementations.

7.3 The Delegation Chain

When AEON-resolved principals invoke agents, the agents inherit a delegated authority from the principal. When agents invoke tools, the tools inherit the same delegation from the agents. When tools invoke other agents or systems, the chain continues. The AEON identity plane is responsible for the integrity of this chain, which means that every action anywhere in the enterprise traces back to a human principal through a verifiable cryptographic chain that AEON itself constructed and maintains.

The chain is constructed at session start. When a principal authenticates through enterprise IDAM, AEON receives the authentication assertion and constructs an authority token that carries the principal's identity, the federation context, the authorized scope, and the expiration. This token is the root of the delegation chain. As the principal interacts with subdomains, AEON issues scoped sub-tokens to the subdomain runtimes, each cryptographically derived from the root token and carrying constrained scope appropriate to the subdomain's authorized operations. As subdomain runtimes invoke agents and tools, they issue further scoped sub-tokens derived from their own sub-tokens. The chain is always traceable backward; every sub-token in the chain can be verified to have descended from a specific principal authentication event.

This chain is what makes agentic accountability possible. Without it, an action somewhere in the enterprise has no architectural mechanism for being attributed to a human principal. With it, every action has a verifiable principal, a verifiable scope, and a verifiable lineage back to the originating authentication event. The principal cannot credibly deny actions her chain authorized; the system cannot credibly attribute actions to principals whose chain did not authorize them.

The chain has to survive subdomain boundaries. When AIDEX agents negotiate with Cybersecurity agents to authorize an operation, when DevSecOps pipelines invoke Digital Engineering models, when Enterprise Platforms workflows touch Insights and Intelligence data, the chain has to compose correctly across the subdomain boundaries. This is the architectural problem that motivates putting identity in the control plane rather than in any subdomain. If the chain belonged to AIDEX, the chain would lose integrity when crossing into DevSecOps; if it belonged to Cybersecurity, the same problem would occur crossing into Enterprise Platforms. The chain has to belong to the architectural layer above all subdomains, which is AEON.

7.4 Revocation and Currency

Identity is not a static fact. Clearances are suspended, program assignments change, separations occur, sessions are terminated. The identity plane has to respond to these state changes with bounded propagation latency, because actions taken under stale identity context are accountability failures.

AEON's revocation propagation operates at two altitudes. At the enterprise IDAM altitude, the source IDAMs publish revocation events to AEON when principal state changes. AEON aggregates these events across the federation and propagates them to subdomain runtimes that hold active sub-tokens for affected principals. The subdomain runtimes are responsible for honoring the revocation: terminating active agents under the affected principal, invalidating outstanding delegations, freezing operations in progress, and closing sessions with whatever cleanup the policy plane specifies.

The propagation latency is a design parameter, not a property of the architecture. Defense contexts typically require near-immediate propagation for high-consequence revocation events, measured in seconds with hard upper bounds enforced by policy. Lower-consequence revocations may propagate over minutes; routine attribute updates may propagate at longer intervals. The propagation tier is set by policy at the authority plane and consumed by the identity plane; the identity plane is responsible for meeting the propagation requirement once the policy is established.

7.5 Degraded Operation

IDAM availability is not guaranteed in defense systems. Networks degrade, services fail, partitions occur. The identity plane has to define behavior when continuous IDAM consultation is not possible. The architectural pattern is time-bounded cached capabilities with cryptographic expiry, established at the identity plane and consumed by the subdomain runtimes.

At session start, AEON issues authority tokens with bounded validity windows. The tokens carry the principal's identity, the delegation scope, the AEON signing context, and the expiry timestamp. Subdomain runtimes can act on these tokens during IDAM unavailability up to the expiry. Beyond the expiry, the subdomain runtime must consult AEON for token renewal, which in turn consults the underlying IDAM. If AEON is unavailable, no renewal occurs and the affected operations terminate.

The validity window is a policy decision per action class. Shorter windows produce more frequent consultation and tighter accountability; longer windows produce more operational resilience and looser accountability. High-consequence action classes get short windows, measured in minutes. Routine action classes can get longer windows, measured in hours or work-shift durations. The window policy is set at the authority plane and enforced by the identity plane through the token expiry construction. Revocation in degraded mode is bounded by the validity window; a revocation event during IDAM unavailability propagates when AEON reaches the affected runtime, which is bounded by the renewal cycle.

7.6 The Subdomain Contract

The identity plane exposes a contract to the subdomains. The contract specifies what the subdomains consume from the identity plane, what the subdomains promise in return, and what the architectural guarantees are.

What the subdomains consume. The identity plane provides principal resolution for any session-initiating event. It provides authority tokens scoped to the subdomain's authorized operations. It provides delegation chain extension when the subdomain spawns agents or invokes tools that require chain continuation. It provides revocation events when principal state changes affect active sessions. It provides federation services when actions cross classification boundaries.

What the subdomains promise. The subdomains promise to bind every action they perform to an AEON-issued authority token. They promise to preserve the delegation chain through their internal call paths, propagating chain context to any agents or tools they invoke. They promise to honor revocation events within their scope, terminating affected operations with the propagation latency policy specifies. They promise to construct chain extensions correctly through AEON-specified protocols rather than reinventing chain mechanics internally.

The architectural guarantees. AEON guarantees that the identity plane is the source of truth for principal identity. AEON guarantees that authority tokens are cryptographically verifiable and tamper-evident. AEON guarantees that the federation broker preserves accountability across classification boundaries. AEON guarantees that revocation propagation meets the latency policy. AEON guarantees that the chain semantics are consistent across subdomain boundaries.

The contract is what makes the identity plane operate as a plane rather than as a service one subdomain happens to use. Every subdomain consumes the same contract; every subdomain's compliance with the contract is verifiable; the architectural integrity of identity across the enterprise depends on contract compliance rather than on any subdomain's internal correctness.

8. The Authority Plane

The authority plane is where AEON answers a question the subdomains cannot answer themselves: is this action authorized? The question has to be answered at the moment of action, against the current attribute state of the principal, with the policy evaluated at enterprise altitude rather than at subdomain altitude. The authority plane owns this answer.

8.1 The Authorization Decision

When a subdomain runtime is about to perform an action on a principal's behalf, it consults the authority plane for an authorization decision. The consultation passes the AEON-resolved principal, the action under consideration, the context in which it would be performed, and any attributes of the artifact or target the action operates on. The authority plane evaluates the request against enterprise policy and returns a decision.

The decision is not a binary allow or deny. The architectural commitment is that the decision space is richer than binary, because enterprise governance over agentic action requires nuance that binary decisions cannot express. The authority plane returns one of seven decision types.

Allow permits the action without modification.

Deny refuses the action and prevents its execution.

Constrain permits the action subject to constraints the subdomain must enforce. The constraint may be on scope (the action may be performed against this subset of data but not the rest), on output (the action may produce this kind of artifact but not that kind), on parameters (the action may use these tools but not others), or on context (the action may be performed in this session but not propagated to others).

Redact permits the action but requires the result to be filtered before it is returned to the principal. Redaction applies when the principal is authorized for the action class but not for some specific content the action would produce.

Route directs the action to be performed by a different mechanism than the one the subdomain originally proposed. Routing applies when policy requires the action to be performed by a more constrained capability, by a different principal authority chain, or in a different execution environment.

Log permits the action with an explicit audit obligation beyond the standard evidence plane capture. Log decisions apply when policy requires special audit treatment for actions that are otherwise permitted.

Escalate requires the action to be confirmed by a higher authority before it executes. Escalation applies when the action is consequential enough that the principal's standing authorization is insufficient and a second authority must approve.

These seven decision types compose. A single authorization decision may include allow with a constraint and a log obligation. The subdomain runtime is responsible for honoring the full decision, not just the allow portion.

8.2 Policy Decomposition

The policy that drives authorization decisions is enterprise-scoped. It is not subdomain-specific. The same enterprise policy that governs how AIDEX handles a sensitive document also governs how Enterprise Platforms processes a related record and how Insights and Intelligence runs analytics on the same data. Policy fragmentation across subdomains produces governance failures; the policy plane prevents fragmentation by being the source of truth.

But enterprise policy cannot be evaluated as a single monolithic ruleset. It has to decompose into subdomain-specific enforcement that the subdomain runtimes can apply efficiently at the moment of action. The decomposition is the architectural challenge of the authority plane.

The pattern is that enterprise policy is authored as policy-as-code in a canonical form (a declarative policy language such as a Rego-class system). The policy plane compiles the enterprise policy into subdomain-specific enforcement bundles that capture the policy provisions relevant to each subdomain's operations. The bundles are deployed to the subdomain runtimes, where they execute against the local action context to produce authorization decisions in milliseconds. The policy plane is responsible for keeping the bundles current as enterprise policy changes.

This pattern is well established in cloud-native infrastructure with OPA-class policy engines. AEON applies the pattern at enterprise altitude with subdomain-specific compilation. The architectural commitment is that policy is authored once at enterprise altitude, evaluated locally at subdomain altitude for performance, and synchronized through AEON's policy distribution mechanism.

8.3 The Compose-and-Evaluate Pattern

Cross-subdomain workflows produce authorization questions that no single subdomain can answer. An AIDEX agent invoking a Cybersecurity scan on a DevSecOps artifact, on behalf of a Digital Engineering principal, produces an authorization question whose answer depends on the composition of authorities across all four subdomains. None of them individually has the policy context to answer.

The authority plane handles this through compose-and-evaluate. When a cross-subdomain workflow proposes an action, the originating subdomain submits the proposal to the authority plane along with the principal's authority context. The authority plane composes the relevant policy across all subdomains involved, evaluates the composition against the action and the principal, and returns a decision that all subdomains can rely on. The composition is the authority plane's responsibility because no subdomain has the architectural standing to perform it.

The composition is not a simple union of subdomain policies. It is an evaluation that considers how the policies interact, what constraints compose across them, what redactions apply across subdomain boundaries, and what escalation paths are appropriate to the cross-subdomain context. The compose-and-evaluate logic is the authority plane's most architecturally substantive responsibility.

8.4 Authority Currency

Authorization decisions are evaluated against the principal's current attribute state. The attribute state can change between when the principal authenticates and when the action is performed. Clearances suspend, program assignments change, role designations update. The authority plane has to evaluate against the current state, not the state at session start.

This produces a continuous-evaluation pattern rather than a cache-at-session-start pattern. The authority plane reads the principal's current attribute state for every consequential authorization decision. For high-frequency low-consequence actions, the authority plane permits short-duration caching based on the subdomain runtime's local enforcement; the cache window is itself a policy decision. For consequential actions, no caching is permitted; the authority plane evaluates against current state every time.

The performance implications are real but not prohibitive. Authority plane consultation operates in the milliseconds for cached attribute reads and the tens of milliseconds for fresh attribute reads. Subdomain runtimes can submit batched consultations for action sequences. The architectural commitment is that current state is evaluated; the optimization patterns are deployment-specific.

8.5 The Behavioral Constraint Layer

The authority plane has a second responsibility beyond per-action authorization. It imposes behavioral constraints on subdomain runtimes that operate orthogonally to principal-based authorization. Some action classes are constrained for the substrate regardless of the principal's authorization. Targeting-adjacent inference, manipulation of certain categories of sensitive data, generation of content in restricted domains, agent autonomy beyond specified bounds, and similar boundaries are constrained at the runtime behavior layer.

These constraints are policy decisions made at the enterprise altitude and enforced by the authority plane through the subdomain runtimes. A principal whose role-based authority would otherwise permit the action cannot direct the substrate to perform it because the constraint operates on substrate behavior, not on principal permissions. The constraint is formal and explicit; it is not an emergent property of training, memory, or model behavior. The authority plane distributes the constraint configuration to subdomain runtimes; the runtimes enforce the constraints regardless of how the principal interacts with them.

The behavioral constraint layer is what allows the enterprise to impose hard limits on agentic action that survive principal pressure. Without this layer, the only way to prevent a substrate from performing a constrained action is to convince the principal not to direct it, which is unreliable. With this layer, the constraint operates at a lower altitude than principal direction; the substrate refuses regardless of what the principal requests.

8.6 The Subdomain Contract

The authority plane exposes a contract to subdomains analogous to the identity plane's contract. Subdomains consume authorization decisions for actions they propose to perform, and they receive behavioral constraint configurations they enforce as runtime properties. Subdomains promise to consult the authority plane before consequential actions, to honor the full decision (including constraints, redactions, routing, and escalation), to apply behavioral constraints regardless of principal direction, and to refuse to perform actions for which they have not received authority plane sanction. The architectural guarantees are that the authority plane evaluates against current policy, that decisions are consistent across subdomains for equivalent actions, that the behavioral constraints are stable in the face of principal pressure, and that the policy decomposition produces enforcement bundles current with enterprise policy.

9. The Evidence Plane

The evidence plane is where AEON makes the enterprise auditable. Every action taken by any subdomain on behalf of any principal contributes to the evidence plane. The contribution captures what happened, who authorized it, what authority decision sanctioned it, what artifacts or side effects resulted, and what context placed the action in sequence. The evidence plane aggregates these contributions into the enterprise audit picture and exposes it for the legitimate purposes audit serves.

9.1 What the Evidence Plane Captures

For every action a subdomain runtime performs, the evidence plane captures the action description in canonical form, the AEON-resolved principal under whose authority it was performed, the authority plane decision that sanctioned it, the subdomain runtime that executed it, any tools or agents the runtime invoked, any artifacts produced, any side effects on systems of record or external systems, the timestamp chain that places the action in sequence with other actions, and any worker confirmation captured at the moment of consequential action.

The capture is structured. Evidence records follow canonical schemas the evidence plane publishes; subdomain runtimes produce evidence in the canonical form rather than in subdomain-specific formats. This is what allows aggregation; evidence from AIDEX, DevSecOps, Cybersecurity, and Enterprise Platforms can be queried together because the records share schema.

The capture is also append-only and integrity-protected. Evidence records cannot be modified after the fact. The append-only commitment is enforced through cryptographic chaining; each record carries a hash that depends on the prior record's hash, producing a chain that any modification would visibly break. The integrity protection is verifiable by any party with access to the chain; tampering cannot be concealed.

9.2 The Aggregation Pattern

Evidence aggregates from subdomain runtimes into AEON's evidence plane through a publish-and-acknowledge protocol. Subdomain runtimes publish evidence records as actions complete, with cryptographic signatures that bind the records to the issuing runtime. AEON's evidence plane acknowledges receipt, verifies the signatures, validates schema compliance, and incorporates the records into the enterprise evidence store.

The aggregation is near-real-time for routine evidence and configurable for higher-volume evidence streams. Some subdomain runtimes produce evidence at rates that warrant batching, sampling, or summarization before submission to the evidence plane; the policy on what to retain at full fidelity versus what to summarize is a policy plane decision consumed by the evidence plane and the subdomain runtimes. The architectural commitment is that the policy is explicit and the evidence treatment under that policy is consistent.

The aggregated evidence is the enterprise audit picture. It is queryable by authorized parties for the purposes audit serves: regulatory compliance audits, internal audit functions, incident response, forensic investigation, dispute resolution, accountability reviews, and the worker-facing access developed in Section 9.5. Each of these query patterns has its own authorization requirements through the authority plane; the evidence plane is the store, the authority plane controls access to it.

9.3 Cross-Subdomain Forensics

The evidence plane's value is largely realized in cross-subdomain forensic analysis. An incident in Enterprise Platforms may have originated in an AIDEX-mediated workflow that touched DevSecOps and Cybersecurity along the way. Reconstructing what happened requires evidence from all four subdomains, correlated through the delegation chain and the action timestamps. The evidence plane is the architectural mechanism that makes this reconstruction possible.

The reconstruction depends on the chain integrity from the identity plane and the canonical schemas the evidence plane publishes. Every action records the delegation chain context in canonical form. Investigators can trace the chain forward from a principal authentication event to identify every action the principal's chain authorized, or backward from an incident to identify the chain of authority that produced it. The forensic queries are expressible against the canonical schema; the queries do not require subdomain-specific knowledge of how each subdomain records its activity.

This is a substantial architectural commitment, because forensic capability across subdomains is exactly what enterprises typically lack when investigating agentic incidents. Each subdomain has its own logs; the logs are not correlated; the incident investigation has to manually reconcile across subdomains, often unsuccessfully. The evidence plane makes the correlation an architectural property rather than a manual reconciliation task.

9.4 Retention and Records Posture

Evidence retention is a policy decision with regulatory, legal, and operational dimensions. Defense contracts impose retention requirements measured in years or decades for some classes of evidence. Privacy regulations impose maximum retention for other classes. Operational practice requires high-resolution evidence for the recent period and summarized evidence for the long tail. The retention policy is set at the policy plane and enforced by the evidence plane through its storage tiering and purge procedures.

The retention policy interacts with the worker-protection commitment developed in Section 9.5 and with the worker contribution revocation question developed in the AIDEX paper. A worker may have a right under enterprise policy to revoke certain contributions she made to institutional practice; the evidence of those contributions and the artifacts they produced may be subject to records retention that overrides revocation. The conflict is real and not resolvable through architecture alone; the architectural commitment is that the conflict is made explicit, that the policy resolves it consistently across subdomains, and that the evidence treatment honors the resolution.

9.5 Worker-Facing Evidence Access

The evidence plane has a worker-protection commitment that is structurally important and frequently overlooked. The evidence the plane captures about a worker's activity is not just an institutional asset; it is the worker's record of what she did under her authority. Workers can query their own evidence to see what their substrate did, what curation they applied, what artifacts resulted from their work, and what audit picture the institution will see if it reviews their activity.

This is consequential for several reasons. It supports the worker's ability to account for her own professional practice when questioned. It allows the worker to detect substrate behavior she did not intend or authorize, before the institution does. It supports the worker in disputes about what she did or did not do under her authority. It builds the trust posture HCAE requires; workers will not commit curation discipline to a system whose evidence they cannot inspect.

The worker-facing access is mediated through the authority plane like all other evidence queries. The worker is authorized to see her own evidence; she is not authorized to see other workers' evidence except where her role and the context warrant it. The architectural commitment is that the worker-facing access exists, is reliable, and is exposed through interfaces the worker can use without specialist support.

9.6 The Subdomain Contract

The evidence plane exposes a contract to subdomains. Subdomains consume the canonical evidence schemas, the publish protocol, the acknowledgment mechanism, and the retention policy distribution. Subdomains promise to publish evidence for every action they perform, to publish in the canonical schema, to bind evidence cryptographically to the issuing runtime, and to handle retention according to the policy the plane distributes. The architectural guarantees are that aggregated evidence is integrity-protected, that cross-subdomain forensic queries are supported by the schema, that retention policy is enforced consistently, and that worker-facing access works without subdomain-specific knowledge of how evidence was originally produced.

10. The Integration Plane

The integration plane is where AEON makes cross-subdomain workflows governable. Subdomains have their own internal orchestrators that compose their internal capabilities into subdomain workflows. The integration plane orchestrates composition across subdomain boundaries, where no single subdomain's orchestrator can preserve the architectural integrity AEON requires.

10.1 Cross-Subdomain Workflow Composition

When a workflow crosses subdomain boundaries, the integration plane is responsible for the composition. Consider a workflow that begins in AIDEX with a principal asking the substrate to assess a system architecture against current security guidance and propose remediations. The substrate may need to retrieve the architecture from Digital Engineering, consult the current security guidance from Cybersecurity, query relevant program context from Enterprise Platforms, and check applicable compliance requirements from Governance and Strategy. The composition of these subdomain interactions into a coherent workflow is the integration plane's responsibility.

The composition is not a simple sequence of calls. The integration plane manages the delegation chain across the subdomain boundaries (the AIDEX agent's authority becomes the basis for the Digital Engineering query, which is verified against the principal's clearance and program assignment from the identity plane). It manages the authorization composition through the authority plane (the composed action is evaluated as a single workflow, not as four independent actions). It ensures the evidence is captured at every step in correlated form. It handles failure modes (what happens if the Cybersecurity query fails halfway through the workflow). It produces a coherent result that the originating AIDEX agent can present to the principal as a single piece of work.

The architectural mechanism is the workflow descriptor. The integration plane accepts a workflow descriptor that specifies the subdomain interactions, the data flow between them, the failure handling, and the result composition. The descriptor is executed by the integration runtime, which invokes the subdomain runtimes through their AEON-specified interfaces, manages the delegation chain, coordinates with the authority and evidence planes, and produces the result. The descriptor itself is an artifact under capability composition governance, addressed in Section 11.

10.2 The Agent Network

The agent network is the specific case of integration that matters most for the agentic era. Agents in one subdomain negotiate with agents in other subdomains on behalf of their principals. An AIDEX agent acting on a principal's behalf may need to invoke a DevSecOps agent that can perform a pipeline operation, which in turn invokes a Cybersecurity agent that can authorize a deployment. The chain of agent-to-agent invocations crosses subdomain boundaries that the agents themselves cannot govern.

The integration plane provides the agent network protocol that handles these invocations. The protocol specifies how agents discover other agents (through the capability registry developed in Section 11), how they authenticate to each other (through delegation chain extension from the identity plane), how they exchange authority (through the authority plane's compose-and-evaluate pattern), how they coordinate (through workflow descriptors that may be implicit in agent-to-agent dialogue), and how they handle failure or refusal.

The protocol commitments are substantial. Every agent-to-agent invocation preserves the principal's identity chain. Every cross-subdomain agent action is evaluated for authorization by the authority plane. Every invocation contributes to the evidence plane. The agents cannot circumvent these by inventing their own inter-agent protocols; the integration plane mandates the protocol they use.

10.3 The Subdomain Adapter Pattern

Subdomains expose their capabilities to the integration plane through adapters. The adapter is a subdomain-side architectural component that translates between the subdomain's internal capability representation and the integration plane's canonical capability contract. The adapter is the responsibility of the subdomain's engineering team, not AEON's team; the adapter contract is defined by AEON and the subdomain implements it.

This pattern is the architectural compromise that lets AEON orchestrate composition without absorbing subdomain operations. The subdomain owns its internal architecture, its runtime, its vendor relationships, its engineering decisions. The subdomain exposes specific capabilities through the adapter in the form AEON specifies. AEON orchestrates the capabilities exposed through the adapter. The subdomain can change its internals freely as long as the adapter contract is preserved; AEON can evolve its orchestration as long as the capability contract is preserved.

The adapters are themselves under capability composition governance. New adapters are registered through the capability registry. Adapter changes go through change governance. Adapter failures are observable through the evidence plane. The adapter pattern is what allows the architecture to be sustained over time without producing brittle subdomain-AEON coupling.

10.4 Composing with Existing Integration Investments

Most enterprises have existing integration infrastructure: enterprise service buses, integration platforms, workflow engines, robotic process automation tools, and various point-to-point integrations accumulated over decades. The integration plane does not replace these. It composes with them under the architectural framework AEON imposes.

The relationship is that existing integration infrastructure becomes a subdomain capability that the integration plane orchestrates, rather than a competitor to the integration plane. An enterprise service bus continues to handle the integrations it was designed for; the integration plane invokes the bus through an adapter when a workflow requires bus-mediated integration. Existing integration platforms continue to run the workflows they were designed for; AEON invokes them when its workflows require composition with them. The integration plane is not a replacement for the integration runtime that already runs in the enterprise; it is the architectural layer that organizes that runtime under enterprise governance.

The architectural commitment is that the integration plane orchestrates composition, not that the integration plane executes every integration. New workflows that benefit from AEON's governance are built natively through the integration plane. Existing workflows are wrapped through adapters when they need to compose with AEON-governed workflows. Workflows that operate entirely within an existing integration platform without composing across subdomains may continue without AEON involvement at all, subject to the policy plane's coverage requirements.

10.5 The Subdomain Contract

The integration plane exposes a contract to subdomains. Subdomains consume the canonical capability registration interface, the workflow descriptor execution interface, the agent network protocol, and the adapter contract that defines how they expose internal capabilities. Subdomains promise to register capabilities through the capability registry, to expose capabilities through compliant adapters, to participate in workflow descriptors as called for by the descriptor's composition, and to honor the agent network protocol for all agent-to-agent invocations. The architectural guarantees are that workflow descriptors execute consistently, that the agent network preserves delegation chain integrity, that the adapter contract is stable over time, and that existing integration investments compose with AEON rather than competing with it.

11. Capability Composition

Capability composition is the architectural function that asks what the enterprise can do, how new capabilities are introduced, how existing capabilities evolve, and how capabilities are retired when they are no longer appropriate. AEON owns capability composition because the answer to those questions is an enterprise property, not a subdomain property. Each subdomain offers capabilities; AEON governs how those capabilities are composed into the enterprise's operational picture.

11.1 The Capability Registry

The capability registry is the canonical record of what the enterprise can do. Every capability any subdomain exposes is registered in the registry along with its metadata: the subdomain that provides it, the capability description in canonical form, the adapter that exposes it, the policy constraints that apply to its use, the lifecycle stage (proposed, qualified, operational, deprecated, retired), the evidence requirements, the audit posture, the dependencies on other capabilities, and the principals authorized to invoke it under what conditions.

The registry is the source of truth for capability discovery. When the integration plane composes a workflow, it queries the registry for the capabilities it can compose. When the authority plane evaluates an authorization decision, it consults the registry for the capability's policy constraints. When the evidence plane categorizes captured evidence, it uses the registry to identify the capability the evidence is about. The registry is consulted by every other plane, which is why it is consolidated at the capability composition altitude.

The registry is not a vendor product. It is a structured store that Enterprise IT builds and maintains because the capability taxonomy is enterprise-specific. Vendors offering "capability catalogs" or "service catalogs" are providing products that may serve as components in the registry's implementation, but the architectural authority over the registry's contents belongs to Enterprise IT.

11.2 The Capability Lifecycle

Capabilities have lifecycles. New capabilities are proposed (typically by subdomain teams, sometimes by enterprise architecture in response to identified needs). Proposed capabilities are qualified through a defined process that evaluates them against architectural fit, policy constraints, evidence sufficiency, and operational readiness. Qualified capabilities become operational and are available for composition. Operational capabilities are monitored for usage, performance, and policy compliance. Capabilities that become obsolete or that present operational concerns are deprecated, transitioned, and eventually retired.

The lifecycle is governed by an explicit process that AEON owns. The process specifies who proposes (subdomain teams, enterprise architecture, others), who qualifies (the AEON team in coordination with the relevant subdomain and the policy plane authorities), what qualification requires (architectural review, policy review, operational review, evidence review), who approves transitions between lifecycle stages, and how deprecation and retirement are communicated to capability consumers.

This is governance work, not just registry maintenance. The AEON team is small and senior; capability lifecycle governance is one of its core responsibilities. The process is deliberate enough to prevent the capability registry from becoming a dumping ground for everything any subdomain wants to call a capability, and lightweight enough that legitimate capability introduction is not strangled by overhead.

11.3 Composition Patterns

Capabilities compose into workflows through patterns the integration plane executes. The composition patterns themselves are first-class artifacts in the architecture; they are not ad hoc combinations of capabilities for each workflow.

Sequential composition invokes capabilities in order, with each capability's output feeding the next. This is the simplest pattern and the most common.

Parallel composition invokes capabilities concurrently and aggregates their results. Common when multiple subdomain queries can be performed independently before synthesis.

Conditional composition selects among capabilities based on runtime conditions, including authority plane decisions. Common when the workflow has multiple legitimate paths depending on principal context or policy.

Iterative composition invokes a capability repeatedly, with adaptation between invocations. Common when the workflow involves refinement or convergence rather than a single pass.

Compensating composition pairs capabilities with their inverses, so that workflows that fail partway through can be unwound coherently. Common when the workflow modifies state in systems of record.

Negotiated composition invokes capabilities through agent-to-agent dialogue rather than predetermined sequences. Common when the workflow's exact composition depends on capabilities discovering and negotiating with each other.

The composition patterns are themselves registered as artifacts under capability composition. New patterns can be introduced when needed; the patterns currently in use are visible to enterprise architecture and policy authorities; pattern usage contributes to the evidence plane.

11.4 Capability Versioning

Capabilities evolve. A capability that was qualified at version 1.0 with one set of policy constraints may evolve to version 2.0 with different constraints. The registry maintains version history. The integration plane resolves workflow descriptor references to specific capability versions. The authority plane evaluates against the policy constraints current for the resolved version. The evidence plane records the capability version invoked.

Versioning matters for several reasons. Workflows that depend on specific capability behavior need version stability over their operating lifetime. Audit posture requires reconstructing what version was in effect at the time of any historical action. Policy evolution requires understanding which versions are subject to which policy variants. The registry maintains versioning as a first-class concern, not as an afterthought.

11.5 Retirement Discipline

Capabilities accumulate. Without explicit retirement discipline, the registry fills with capabilities that are no longer appropriate, no longer supported, no longer secure, no longer aligned with policy, but still technically available because no one has retired them. This is the same accretion problem that produces dead code in software, dead processes in business operations, and dead artifacts in document repositories. The registry has to be actively curated to avoid the same fate.

The retirement discipline is part of the lifecycle governance. The AEON team reviews the registry on a defined cadence, identifies capabilities that are candidates for deprecation, coordinates with the providing subdomains and the consuming workflows, and executes the retirement when the conditions are met. The discipline requires institutional commitment; an AEON team that prioritizes adding capabilities over retiring them produces a registry that grows linearly without growing in quality.

11.6 The Subdomain Contract

The capability composition plane exposes a contract to subdomains. Subdomains consume the capability registration interface, the lifecycle process for moving their capabilities through stages, the composition pattern definitions, and the versioning conventions. Subdomains promise to register capabilities through the canonical interface, to participate in the lifecycle process for their capabilities, to maintain version discipline, and to honor retirement decisions. The architectural guarantees are that the registry is the source of truth, that the lifecycle process is consistent across subdomains, that composition patterns are stable artifacts, and that retirement happens deliberately rather than through neglect.

12. The Meta-Orchestration Runtime

The meta-orchestration runtime is what Enterprise IT actually builds, deploys, and operates. The first eight sections specified what AEON does as an architecture. This section specifies what AEON is as a running platform. The distinction matters because architectural specifications can be elegant without being buildable; the runtime specification is where architecture meets engineering reality.

12.1 What the Runtime Is

The AEON runtime is a coherent platform that hosts the planes (identity, authority, evidence, integration, capability composition) as services that subdomain runtimes consume. The runtime is not the planes themselves; the planes are architectural service definitions that the runtime implements. The runtime is the engineered platform that exposes the plane interfaces, executes the plane logic, integrates with the underlying components Enterprise IT has selected, and provides the operational backbone the control plane requires.

The runtime is built by Enterprise IT as a composition of components. The identity plane is implemented through integration with the enterprise IDAM, a federation broker Enterprise IT builds or selects, and the token issuance and chain management services Enterprise IT operates. The authority plane is implemented through a policy engine Enterprise IT selects (an OPA-class policy-as-code system, or equivalent), the policy compilation pipeline Enterprise IT builds, and the policy distribution and consultation services Enterprise IT operates. The evidence plane is implemented through an evidence store Enterprise IT selects (layered over the existing SIEM and audit stack, or equivalent), the schema enforcement services Enterprise IT builds, and the query and access services Enterprise IT operates. The integration plane is implemented through a workflow runtime Enterprise IT builds or selects, the agent network protocol Enterprise IT specifies and operates, and the adapter contracts Enterprise IT enforces. Capability composition is implemented through a registry Enterprise IT builds, the lifecycle process Enterprise IT operates, and the pattern definitions Enterprise IT curates.

The runtime is a substantial piece of engineering. It is not as substantial as building the enterprise IDAM from scratch, the SIEM from scratch, or the policy engine from scratch, because those are vendor-supplied components. It is more substantial than the typical enterprise integration project because the architectural commitments are stronger and the scope is enterprise-wide. The runtime is Enterprise IT's program for the next several years, not a single project.

12.2 Deployment Topology of the Runtime

AEON itself is deployed across the classification environments the enterprise operates in. There is not one AEON; there are coordinated AEON instances, one per classification environment, with explicit federation between them. The federation is what Section 14 develops in detail; here the architectural point is that the runtime is itself a federated platform.

The unclassified AEON instance hosts the identity plane that resolves unclassified principals, the authority plane evaluating unclassified policy, the evidence plane aggregating unclassified evidence, and the integration plane orchestrating unclassified workflows. The CUI AEON instance does the same for CUI scope. The classified AEON instance does the same for classified scope, and may itself be federated across program boundaries. Each instance is operationally autonomous within its scope; the federation between instances is what allows principals, policies, and evidence to compose across the classification boundary in the controlled ways the enterprise's posture permits.

The deployment within each instance is conventional enterprise platform architecture. The runtime services are deployed on infrastructure the Infrastructure and Operations subdomain provides. The services are containerized, orchestrated through standard container orchestration platforms, monitored through standard observability infrastructure, secured through standard enterprise security posture. The runtime is not exotic infrastructure; it is conventional enterprise infrastructure running services that implement the AEON architecture.

12.3 Performance Characteristics

The runtime has to meet performance requirements that subdomain runtimes can plan against. Authorization plane consultation should return decisions in tens of milliseconds for fresh evaluations and single-digit milliseconds for cached evaluations. Identity plane token issuance should complete in tens of milliseconds. Evidence plane publication should be near-real-time for routine evidence with bounded backpressure when volume spikes. Integration plane workflow execution should add minimal overhead compared to direct subdomain calls.

These targets are achievable with the conventional platform architecture. They are not free; the runtime has to be engineered for them. Caching at appropriate altitudes, asynchronous publication where consistency requirements allow, deployment of services close to the subdomain runtimes that consume them, and the standard engineering disciplines that mature platform teams practice are all necessary. The performance characteristics are themselves part of the architectural contract with subdomains; the contract is unenforceable if the runtime cannot meet it.

12.4 Operational Posture

The runtime is operated like a critical enterprise platform. It is monitored continuously. Its operational state is itself reflected in the evidence plane (the platform that captures enterprise evidence also captures evidence about itself). Its degraded states are explicit and documented. Its failure modes are tested through chaos engineering practice. Its recovery procedures are exercised. Its capacity is planned against subdomain growth projections. Its security posture is part of the Cybersecurity subdomain's purview, even though the platform itself is operated by the AEON team.

The operational practice matures over time. Early in the deployment, the runtime is operated cautiously with conservative defaults and substantial manual oversight. As confidence develops and operational experience accumulates, defaults loosen, automation expands, and manual oversight focuses on exceptions rather than routine operations. This maturation is normal for platform engineering at this scope; the AEON team plans for it explicitly rather than expecting to operate at maturity from day one.

12.5 Observability of the Runtime Itself

The runtime is observable. Every plane exposes metrics about its operation: authorization decisions per second, identity resolutions per second, evidence records published per second, workflow executions per second, capability registry operations per second. Every plane exposes its operational state: healthy, degraded, failed, in maintenance. Every plane exposes its capacity utilization. The metrics and state feed into enterprise observability infrastructure that the AEON team and the broader Enterprise IT function use to operate the platform.

The observability is also exposed to the subdomain teams that depend on AEON. They can see the operational state of the planes they consume. They can plan around degraded states. They can correlate their own operational issues with AEON operational issues. The transparency is part of the architectural commitment; AEON is not a black box the subdomains query without insight into its state.

13. The Enterprise IT Operating Model for AEON

AEON requires a specific Enterprise IT function to build and sustain it. The function is not the same as the function that ran the pre-agentic Enterprise IT. The capabilities are different, the operating model is different, the relationship to the rest of Enterprise IT is different. This section specifies what the function looks like when it is constituted correctly.

13.1 Function Scope and Authority

The AEON function builds and operates the AEON architecture. Its scope is the control plane: the planes, the runtime, the operating model, the evolution. Its authority is architectural authority over the control plane and over the contracts subdomains consume. It does not have authority over subdomain internals; subdomain teams own their internal architecture and operations. It does have authority over the contracts subdomains expose through their adapters, because the contracts are what make the architecture coherent.

The function reports within Enterprise IT, typically to a Chief Architect or equivalent senior role. It is positioned at the same altitude as the enterprise architecture function in mature enterprises; in some organizational structures, it is part of the enterprise architecture function. The reporting relationship matters because the function's authority depends on it; an AEON team buried under a delivery organization will not have the architectural standing to govern subdomain contracts.

The function does not have authority over subdomain technology selection, vendor relationships, or operational decisions. The Digital Engineering subdomain selects its modeling tools; the Enterprise Platforms subdomain selects its ERP; the Cybersecurity subdomain selects its security tooling. AEON's authority is over how these subdomains compose into the enterprise's operational picture, not over what they choose internally.

13.2 Team Composition and Skills

The AEON team is small and senior. Small because the architectural authority it exercises depends on coherence that a large team would struggle to maintain. Senior because the work requires architectural judgment that does not develop through years of conventional Enterprise IT practice; it requires explicit cultivation of control plane thinking.

The skills required are several and not entirely conventional. Identity engineering at the level the identity plane requires is a specialized discipline. Policy engineering at the level the authority plane requires is similarly specialized. Evidence engineering for agentic action is a discipline that is still being invented across the industry. Integration architecture at the meta-orchestration altitude is rarer than integration engineering at the platform altitude. Capability composition as an architectural practice has very few practitioners with substantial experience. The team has to be assembled from people with these skills, often recruited from outside the enterprise's current Enterprise IT function.

A reasonable team size is small dozens, not hundreds. The skill density is what matters, not the headcount. A team of fifteen senior architects and engineers with the right skills produces more than a team of fifty conventional Enterprise IT staff who do not have control plane experience. The institutional commitment is to that skill density, even when the cost per person is higher than conventional Enterprise IT roles.

13.3 Relationship to Subdomain Engineering Teams

The AEON team's relationship with subdomain engineering teams is collaborative but architecturally asymmetric. The AEON team specifies the contracts; the subdomain teams implement them. The AEON team participates in subdomain capability qualification; the subdomain teams build and operate the capabilities. The AEON team operates the control plane runtime; the subdomain teams consume it.

The collaboration takes the form of joint architecture reviews, contract development through partnership rather than dictation, capability lifecycle management as a shared process, and ongoing operational coordination as the platforms evolve together. The architectural asymmetry is that when contract decisions are required, the AEON team has the deciding authority; when subdomain operational decisions are required, the subdomain team has the deciding authority. Neither can override the other within the other's scope.

This relationship requires institutional support. Subdomain teams accustomed to autonomy may resist the AEON contracts; AEON teams without institutional backing may be unable to enforce the contracts. The Chief Architect role that the AEON function reports to, or an equivalent senior authority, has to provide the institutional backing that makes the architectural authority real. Without that backing, the architecture is unenforceable and the enterprise gets the accidental composition the architecture was designed to prevent.

13.4 The Architecture-as-Code Practice

The AEON team operates the architecture as code. The plane contracts, the policy decompositions, the workflow descriptors, the composition patterns, the capability registrations are all expressed in code or code-adjacent declarative form and managed through software engineering practice. They are version-controlled. They are reviewed. They are deployed through pipelines. They are tested. They are documented.

This is not how all enterprise architecture functions operate. Many operate through documents, PowerPoints, governance reviews, and architecture review boards. The AEON function cannot operate that way because the architecture has to execute at runtime; documents that are not executable cannot constrain runtime behavior. The architecture-as-code practice is a precondition for the architecture being real rather than aspirational.

The practice has implications for the team's skill mix. The team needs people who can write code, not just architects who can produce diagrams. The team needs DevSecOps maturity, not just architecture review maturity. The team needs to operate its own infrastructure as code, not just specify infrastructure for others to operate. The skill mix is part of what makes the team's composition different from conventional enterprise architecture functions.

13.5 Governance Cadences

The AEON function operates several governance cadences that the rest of Enterprise IT participates in. Architecture review boards evaluate contract changes, new capabilities entering qualification, policy plane changes that affect subdomain behavior, and the like. Operational reviews assess the runtime's state, its capacity, its incident posture, its evolution. Strategic reviews assess the architecture's alignment with enterprise direction, its response to industry shifts, its prioritization of investment.

The cadences are deliberate. Weekly operational reviews, monthly architecture review boards, quarterly strategic reviews are reasonable defaults. The participants vary by cadence: operational reviews are internal to the AEON team and immediate stakeholders; architecture review boards include subdomain engineering leads and enterprise architecture; strategic reviews include senior Enterprise IT leadership and business representation.

The cadences produce decisions and they produce evidence. Decisions are captured in the architecture-as-code artifacts; evidence is captured in the evidence plane (the operational reviews about the runtime are themselves audited like any other action). This produces a self-documenting governance function that survives staff turnover and supports institutional learning.

13.6 The Capability to Build the Function

Most enterprises do not currently have the function this section describes. They have Enterprise IT functions organized around application delivery, infrastructure operations, and platform engineering, with enterprise architecture as an advisory function. Building the AEON function from this starting point is itself a substantial program.

The pattern is to start with a senior architect with appropriate background, give the person authority and resources, and build the team deliberately around them. The initial team can be six to ten people focused on the most foundational planes (identity and authority). Additional capability is added as the team demonstrates that the early planes are operational and trusted. The team’s growth is paced by the maturity it demonstrates rather than by a predetermined plan.

The institutional commitment is significant. The enterprise commits to recruiting senior talent that may cost more than conventional roles. It commits to giving the function architectural authority that may produce friction with existing teams. It commits to sustaining the function over a multi-year program rather than expecting outcomes in a single budget cycle. Enterprises not willing to make these commitments should not start the program; partial commitment produces partial AEON, which is worse than not having AEON at all.

14. AEON Across Classification Environments

Defense enterprises operate across multiple classification environments. The architecture has to handle this without producing fragmentation that defeats the purpose of having an enterprise control plane.

14.1 The Federated AEON Instance Pattern

AEON deploys as coordinated instances across the classification environments. There is an unclassified AEON, a CUI AEON, a NOFORN or restricted AEON where the enterprise’s operations require it, and a classified AEON or multiple classified AEONs at higher levels. Each instance is operationally autonomous within its environment. Each instance has its own identity plane integrating with the IDAM appropriate to its environment, its own authority plane evaluating policy appropriate to its scope, its own evidence plane aggregating evidence for its environment, its own integration plane orchestrating workflows within its scope, its own capability registry of capabilities appropriate to its environment.

The instances are coordinated through federation rather than being a single distributed AEON. The federation respects the classification posture: information flows from higher classification to lower classification under explicit policy controls, never the reverse for sensitive information, with cross-domain mechanisms appropriate to the boundary.

The architectural commitment is that the instances follow the same architecture. The plane contracts are identical across instances. The runtime is the same platform engineered for the classification context. The operating model is the same function staffed at each environment. The federation between instances follows defined patterns rather than being ad hoc.

14.2 Identity Federation

Identity federation across AEON instances follows the patterns developed in Section 7.2. Derived identities establish that a principal authenticated in one environment has a verifiable corresponding identity in another. One-way attribute synchronization propagates principal state from higher to lower classification environments. Portable capability tokens allow specific authorized operations to cross boundaries. Cross-domain identity brokers mediate the federation under policy control.

The federation has direction. Information about principal existence may flow downward (a principal who exists in a NOFORN environment may have a derived identity in the CUI environment if policy allows). Information about principal classification status flows upward only under explicit policy and only in summarized forms (the CUI AEON may know that some of its principals also have classified standing without knowing the specifics). The directionality is enforced at the federation broker, not by the goodwill of any individual operator.

14.3 Policy Federation

Policy at the enterprise altitude often spans classification environments. The same enterprise policy that governs a category of action in unclassified contexts also governs it in CUI and classified contexts, with appropriate refinements. The authority plane has to handle this without producing policy drift across instances.

The pattern is that policy is authored at the highest classification context where the policy is needed and propagates downward to lower classification instances along with the federation flow. Each AEON instance receives the policy current for its scope; instances at higher classification have access to the full policy; instances at lower classification have the policy provisions appropriate to their scope. Policy changes propagate from higher to lower with explicit publishing actions rather than implicit synchronization.

Some policy is environment-specific. NOFORN environments have policies that do not exist in unclassified environments and would not be appropriate there. The architecture supports this: the authority plane in each instance may have policy that exists only in that instance, while sharing the policy that crosses environments. The composite policy picture is consistent within each instance but not identical across them.

14.4 Evidence Federation

Evidence federation across AEON instances follows the directional pattern. Evidence from higher classification instances does not flow downward in detail; lower classification instances may receive summaries or counts of activity at higher classification, but not the specific evidence records. Evidence from lower classification instances may flow upward to higher classification under policy that allows it, when the higher-classification context has legitimate audit interest in lower-classification activity.

This pattern is necessary because the evidence itself may carry classified information; an evidence record from a classified AEON describing what a principal did contains the action description, which may itself be classified. The federation has to preserve classification while still supporting cross-instance audit picture for enterprise audit purposes that have legitimate cross-instance scope.

The enterprise audit picture, when cross-instance scope is required, is constructed through queries that span instances under explicit authorization. The query is evaluated in each instance against the instance's policy and evidence; the results are composed at the query level rather than at the storage level. This avoids producing a unified evidence store at the highest classification, which would defeat the federation's purpose.

14.5 Cross-Domain Composition

Some workflows legitimately need to compose across classification environments. A principal working in CUI may need to reference unclassified material; an action taken in classified context may need to consult an unclassified policy reference; a workflow that operates primarily in CUI may need a step that the unclassified environment can perform.

The integration plane handles this through cross-domain composition patterns that respect the federation directionality. A workflow can reach down from CUI to unclassified for unclassified material under policy that allows it. A workflow cannot reach up from unclassified to CUI for CUI material, regardless of policy; the unclassified context does not have the architectural means to invoke CUI operations. The asymmetry is enforced through the architecture, not through policy alone.

The cross-domain composition patterns are themselves under capability composition governance. They are registered, qualified, monitored, and retired like other capabilities, with the additional discipline that cross-domain capability changes require coordination across the federation rather than within a single instance.

14.6 Coordinated Operations

The AEON function operates the federated instances coordinately. The function in each environment may be staffed separately if the classification posture requires it; the architectural decisions and the operating model are coordinated across the function. Changes to plane contracts propagate across all instances. Changes to capability lifecycle process apply consistently. Changes to runtime engineering apply across all instances with appropriate classification-specific adjustments.

The coordination is governance work. The senior AEON architect role, or its equivalent, coordinates across the instances. Architecture review boards include representation from each instance. Strategic reviews assess the federation as a whole. The coordination produces a coherent enterprise AEON rather than disconnected per-environment platforms that happen to share an architectural pattern.

15. Composing with Existing Investments

AEON is not a greenfield architecture. It is an architectural overlay that organizes existing enterprise investments while adding what is missing. The relationship to existing investments determines whether AEON is buildable in an enterprise that already has significant infrastructure.

15.1 The Compositional Posture

The compositional posture established in Section 6.2 applies throughout. AEON composes with existing enterprise investments rather than replacing them. The existing IDAM is the foundation of the identity plane, with the AEON federation broker overlaying it. The existing SIEM and audit infrastructure is the foundation of the evidence plane, with the AEON canonical schema overlay providing the cross-subdomain queryability. The existing integration platforms are subdomain capabilities the integration plane orchestrates, with new cross-subdomain workflows built natively through AEON. The existing policy infrastructure, where it exists, integrates with AEON's authority plane.

The compositional posture means that AEON deployment does not require the enterprise to throw away current infrastructure. The current infrastructure is mostly preserved, augmented by the AEON architecture, and over time evolved under the architecture's guidance.

15.2 The Identity Investment Inventory

Most enterprises have substantial existing identity infrastructure. Federated identity providers (Active Directory Federation Services, Okta, Ping, equivalent), specific identity stores for application access, role-based access management systems, attribute provisioning pipelines, and various identity governance and administration tools. The AEON identity plane composes with this inventory.

The federation broker is the new component. It mediates between existing IDAM instances and exposes the AEON identity plane contract to subdomain runtimes. The existing IDAMs continue to perform authentication and to be the source of truth for their respective principal populations. The federation broker translates between them and provides the unified identity surface AEON requires.

In most enterprises, this means the existing IDAM investment is preserved and the AEON layer is built on top of it. In some enterprises, the existing IDAM is fragmented enough that consolidation or rationalization is itself a prerequisite for AEON; the federation broker can mediate between a few IDAMs but not between dozens. The IDAM rationalization work, where required, is a precursor to AEON deployment rather than something AEON itself addresses.

15.3 The Policy Investment Inventory

Policy infrastructure varies more widely across enterprises than identity infrastructure does. Some enterprises have mature policy-as-code practice with explicit policy engines deployed in production; others have policy expressed primarily in documents and human governance processes with little or no runtime enforcement; most are somewhere in between.

The authority plane composes with whatever exists. Where policy-as-code practice exists, the authority plane adopts and extends it; the policy engine becomes part of the authority plane's implementation. Where policy lives primarily in documents, the authority plane introduces policy-as-code as a new practice, starting with the policies most directly relevant to agentic action and expanding from there. The introduction is itself a substantial program; policy translation from documents to code is not mechanical and requires legal, compliance, and operational engagement.

The architectural commitment is that the authority plane evaluates policy as code at runtime. Whatever exists today that does not evaluate as code at runtime is not authority plane infrastructure yet; it is policy that needs to be translated before it can be enforced architecturally. This translation work is one of the most substantial Enterprise IT investments AEON deployment requires.

15.4 The Audit and Evidence Investment Inventory

Most enterprises have substantial audit and evidence infrastructure. SIEMs (Splunk, Sentinel, Elastic, equivalent), audit log management, compliance reporting platforms, forensic investigation tools. The AEON evidence plane composes with this inventory.

The canonical schema is the new commitment. Existing audit infrastructure typically captures evidence in tool-specific formats that do not compose across the inventory. The AEON evidence plane introduces canonical schemas that subdomain runtimes produce; the existing audit infrastructure is augmented to ingest, store, and query against the canonical schemas in addition to its existing capabilities.

The compositional pattern is to deploy the canonical schemas progressively. The most consequential subdomain runtimes for agentic action (typically AIDEX and Cybersecurity) adopt the schemas first; other subdomains adopt them as they enter the AEON architecture. The existing audit infrastructure continues to operate; the canonical schemas overlay it without replacing it.

15.5 The Integration Investment Inventory

Enterprise integration architecture is typically the most layered of the existing investments. Enterprise service buses, integration platforms, workflow engines, robotic process automation, point-to-point integrations, ETL pipelines, and various flavors of API gateway and message broker accumulate over decades. The AEON integration plane composes with all of it.

The pattern is the subdomain adapter. Existing integration capabilities are exposed to AEON as subdomain capabilities through adapters. The existing integration runs unchanged; the adapter exposes specific operations to the integration plane for use in cross-subdomain workflows. New workflows that benefit from AEON governance are built natively through the integration plane; existing workflows are wrapped through adapters when they need to compose across subdomain boundaries; workflows that operate within a single subdomain may continue without AEON involvement.

This produces a layered integration architecture in which AEON's integration plane sits above the existing integration investments and orchestrates them where cross-subdomain composition is required. The existing investments are preserved and continue to deliver their existing value; AEON adds the cross-subdomain composition that existing investments cannot provide.

15.6 The Platform Investment Inventory

Subdomain platforms (ERP, PLM, CRM, engineering data management, security operations platforms, DevOps tooling, data and analytics platforms) are the operational backbone of the existing enterprise. AEON composes with all of them through the adapter pattern developed in Section 10.3. The platforms continue to operate; the adapters expose specific capabilities to the integration plane; the platforms compose into cross-subdomain workflows when needed.

The platforms are not part of AEON. They are subdomain capabilities that AEON orchestrates. The distinction matters because it preserves the subdomain teams' authority over their platform decisions; AEON does not select the ERP, the PLM, or any other platform. AEON specifies how the platforms expose capabilities to the enterprise composition layer; the platform selection itself remains with the subdomain.

15.7 The Migration Sequence

Existing investments are not migrated to AEON all at once. The compositional posture means that most existing investments are not migrated at all; they are composed with rather than replaced. The migration that does occur is selective and sequenced.

Identity federation broker deployment is typically the first migration step. Without it, the identity plane cannot provide its services to subdomain runtimes. Policy compilation pipeline deployment is typically the second step, starting with a small set of high-priority policies and expanding from there. Canonical evidence schema adoption is sequenced by subdomain, starting with the subdomains most consequential to agentic action. Integration plane workflow deployment is sequenced by use case, starting with cross-subdomain workflows that have clear value and expanding as adoption builds. Capability registry deployment is foundational and happens early, with the capability inventory built incrementally as subdomains expose their capabilities.

The migration sequence is one of the most consequential strategic decisions in AEON deployment. Too aggressive a sequence produces deployments that fail because the existing investments cannot absorb the architectural overlay quickly enough. Too cautious a sequence produces deployments that do not deliver value quickly enough to sustain institutional commitment. The right pace is enterprise-specific and develops through experience; the architecture supports a range of paces rather than imposing a single one.

16. Phased Deployment and Sustained Evolution

The AEON deployment is a multi-year program. It is not a project with a defined endpoint; it is an institutional capability that the enterprise commits to building and sustaining indefinitely. The phased deployment establishes the capability; the sustained evolution maintains and grows it.

16.1 The Minimal Coherent Subset

Before describing the full phasing, it is worth naming the smallest version of AEON that delivers value on its own. The full six-plane architecture is the end-state, but the enterprise does not have to reach the end-state to benefit. There is a minimal coherent subset that delivers real governance value while requiring far less than the full architecture, and treating AEON as a sequence of valuable plateaus rather than an all-or-nothing bet reduces the perceived risk of starting.

The minimal coherent subset is the identity plane, the authority plane, and a minimal evidence plane, supporting one or two high-value cross-subdomain workflows. Concretely, the smallest version worth deploying is AEON governing the interaction between AIDEX and Cybersecurity: the identity plane resolving principals and maintaining the delegation chain between the worker substrate and the security subdomain, the authority plane evaluating whether the substrate's proposed actions are authorized under enterprise security policy, and a minimal evidence plane capturing what the substrate did under whose authority with what security sanction.

This subset delivers value that stands on its own regardless of whether the enterprise ever builds the rest. It provides enforced human accountability for the worker substrate, which is the single most consequential governance gap in early agentic adoption. It provides agentic guardrails at the security altitude, because the authority plane's behavioral constraint layer can impose hard limits on substrate behavior. It provides forensic capability for the substrate's activity, because the minimal evidence plane captures the substrate's actions in correlated form. An enterprise that deploys only this subset has solved the most urgent agentic governance problem, even though it has not yet built the integration plane, the full evidence plane, capability composition, or the broader subdomain coverage.

The minimal subset is also the natural proving ground for the function. It exercises the hardest planes (identity and authority) at limited scope, lets the function develop operational maturity before broadening, and demonstrates value early enough to sustain institutional commitment for the fuller program. The phases that follow build outward from this subset rather than toward a distant end-state that delivers nothing until it is complete.

16.2 Phase Zero: Function Establishment

Before AEON is built, the AEON function has to exist. Phase zero is the establishment of the function: recruiting the senior architect to lead it, building the initial team of six to ten people with the right skills, securing the institutional authority the function requires, defining the program scope, securing the budget for multi-year work.

Phase zero is not deployment work in the technical sense. No platform is being built yet. But phase zero work is foundational; an AEON function that begins without the institutional grounding will not succeed in the deployment phases. Phase zero typically takes six to twelve months, depending on the enterprise's starting state and the difficulty of recruiting the senior talent the function requires.

16.3 Phase One: Foundational Planes

Phase one builds the foundational planes: identity and authority. These two planes are the foundation that every other plane and every subdomain depends on. They have to be built first, deployed first, and stabilized first before later phases proceed.

Phase one work includes the identity federation broker deployment, the policy compilation pipeline deployment, the initial set of policies translated into policy-as-code, the integration of the planes with one or two pilot subdomains (typically Cybersecurity and AIDEX, because they are most consequential to agentic action), and the operational maturity to run the planes reliably.

Phase one typically takes twelve to eighteen months from a phase zero baseline. The end of phase one is a working identity plane and authority plane that the pilot subdomains depend on for their operation, with the runtime stable enough to broaden to additional subdomains.

16.4 Phase Two: Evidence and Integration

Phase two adds the evidence plane and the integration plane. With identity and authority foundational, the evidence plane begins capturing canonical evidence from the pilot subdomains, and the integration plane begins orchestrating cross-subdomain workflows.

Phase two work includes the canonical evidence schema rollout, the evidence aggregation runtime, the worker-facing evidence access, the workflow descriptor runtime, the agent network protocol implementation, the initial set of cross-subdomain workflows built natively through AEON, and the operational maturity to handle the additional planes' load.

Phase two typically takes twelve to eighteen months from a phase one baseline. The end of phase two is a working evidence plane and integration plane, with the pilot subdomains contributing evidence and participating in cross-subdomain workflows, and the architecture providing the cross-subdomain forensic capability that distinguishes AEON from pre-AEON enterprise architecture.

16.5 Phase Three: Capability Composition and Broader Adoption

Phase three formalizes capability composition and expands AEON to additional subdomains. The capability registry, the lifecycle process, the composition patterns, and the broader subdomain enrollment happen in phase three.

Phase three work includes the capability registry deployment, the lifecycle process establishment and operation, the composition pattern definitions, the enrollment of additional subdomains beyond the pilots, and the operational expansion to handle enterprise-wide load. The phase also includes the early classification federation work, deploying coordinated AEON instances across the classification environments the enterprise operates in.

Phase three typically takes eighteen to twenty-four months from a phase two baseline. The end of phase three is a working AEON that covers the major subdomains, with the federation across classification environments at least partially operational, and the architecture established as the enterprise's control plane.

16.6 Sustained Evolution

Beyond phase three, AEON enters sustained evolution. The architecture continues to develop as the enterprise's agentic capability grows. New planes may be added; existing planes evolve; capability composition matures; the operating model refines. The function continues to operate the runtime, evolve the architecture, and support the subdomain teams.

The sustained evolution is the steady state. It is where AEON delivers its value over the long run. The early phases establish the architecture; the sustained evolution maintains it and grows it under coherent architectural governance.

16.7 What Could Go Wrong

The phased deployment has predictable failure modes that the program has to design against.

Phase zero failure occurs when the function is established without sufficient institutional commitment. The senior architect is hired but not given authority. The team is funded but not at the seniority required. The program scope is approved but the budget is single-year rather than multi-year. The function fails before it builds anything because it does not have the foundation for the work.

Phase one failure occurs when the foundational planes are attempted before the function is mature. Identity engineering proves harder than expected. Policy translation runs into legal and compliance complications. The pilot subdomains do not engage adequately. The runtime is engineered too quickly to be stable. The phase ends with planes that nominally exist but do not work reliably enough to support phase two.

Phase two failure occurs when the integration plane is attempted before the foundational planes are stable. Cross-subdomain workflows fail because identity does not flow correctly. Evidence aggregation fails because the canonical schemas are not adopted consistently. The integration plane reveals weaknesses in the planes below it that should have been addressed before integration.

Phase three failure occurs when capability composition is attempted before the planes below it are operating reliably. The capability registry becomes a dumping ground because the lifecycle process is not operated rigorously. The composition patterns proliferate without architectural discipline. The subdomain enrollment outruns the operational capacity of the AEON function.

Sustained evolution failure occurs when the function loses institutional commitment over time. Budgets are cut. Senior talent leaves and is not replaced. The architecture stagnates while subdomain demands grow. The enterprise's control plane gradually erodes back to the pre-AEON pattern through neglect.

These failure modes are real and worth designing against. The countermeasures are institutional rather than technical: secure the institutional commitment before the program begins, pace the phases to the function's maturity, sustain the commitment beyond the initial deployment.

17. Open Questions, Concerns, and Opportunities

The AEON architecture is mature enough to commit to and detailed enough to build, but several questions remain open and several concerns warrant explicit acknowledgment.

17.1 Open Questions

Cross-vendor capability composition. The architecture assumes subdomain capabilities can be composed through standard adapters. Vendor capabilities vary in how cooperatively they expose adapter-friendly interfaces. Some vendors resist standardized contract exposure because it weakens their lock-in posture. The architectural response is to require the capability through procurement; the practical response is to discover during deployment which vendors can be composed cleanly and which cannot.

Performance at scale. The runtime performance characteristics specified in Section 12.3 are achievable at moderate scale. Whether they hold at the largest defense enterprises with hundreds of thousands of users and many thousands of subdomain runtime instances is not yet established. The architecture provides scaling patterns; the engineering proof points accumulate during deployment.

Policy translation completeness. Translating enterprise policy into policy-as-code is one of the harder elements of phase one work. The translation is not always tractable; some policies depend on contextual judgment that does not compile cleanly into runtime evaluation. The architectural response is the seven decision types and the escalate option for actions that exceed compilable policy. The practical question is how much policy can be compiled and how much remains in escalation.

Capability composition governance overhead. The capability lifecycle process has to be rigorous enough to prevent registry decay and lightweight enough to support legitimate capability introduction. The right balance is enterprise-specific and evolves with the function's maturity. There is no single optimal answer.

Cross-domain federation completeness. The federation patterns developed in Section 14 are sound architecturally. Whether they have been exercised at scale in defense environments with the specific classification topologies any individual enterprise operates is not certain. The patterns require validation in actual deployment.

The AEON instance topology under M&A. Defense enterprises acquire and divest other organizations. Each acquisition brings its own enterprise architecture, including possibly its own AEON instance or pre-AEON infrastructure. The architectural patterns for absorbing acquired organizations into the federation are not fully developed; this is work the architecture has to mature into.

The relationship to vendor "agentic orchestration" platforms. The architecture explicitly rejects vendor-supplied control planes. As vendors increasingly market "agentic AI platforms" that claim to be the control plane, the strategic argument for enterprise-owned AEON has to be communicated and held against the institutional pressure to outsource the work. The strategic question is whether enterprises hold the line.

Open-source contribution posture. Enterprise IT-built does not preclude contributing to and benefiting from open-source software in the AEON stack. The question is which components are appropriate to develop openly versus which remain enterprise-internal. This is institutional posture work rather than architectural work, but it affects how the function is staffed and operated.

17.2 Concerns

Vendor capture by stealth. The compositional architecture depends on vendor components being replaceable. Vendors have strong incentives to make their components less replaceable than the architecture assumes. Capture happens through specialized features that get adopted without architectural review, through performance tuning that makes alternative components hard to substitute, through pricing structures that lock the enterprise in incrementally. The function has to actively defend the compositional architecture against capture; the defense is ongoing rather than one-time. The concrete mechanisms are procurement guardrails: contract language that mandates adapter-friendly APIs exposing the capability through AEON's contract, explicit non-exclusive control-plane clauses that prohibit the vendor from claiming the orchestration layer, data and configuration portability clauses that guarantee the enterprise can extract and migrate, and architectural review as a gate on adopting vendor-specific features that would deepen coupling. The guardrails go into the procurement template so they apply to every acquisition rather than depending on case-by-case vigilance.

Institutional commitment erosion. AEON requires sustained commitment over many years. Budget cycles, leadership transitions, strategic pivots, and competing priorities all create pressure to redirect investment from sustained infrastructure work to more visible delivery work. The function's value is largely invisible when it works well; failures get attention, smooth operations do not. The concrete mechanisms make abandonment costly and deliberate rather than a matter of drift: bind AEON milestones into the multi-year portfolio roadmap so they are tracked as committed deliverables, enter the dependency of agentic governance on AEON into the enterprise risk register so that defunding AEON requires explicit risk acceptance signed at the appropriate level, and report AEON's governance outcomes (audit time-to-investigate, compliance certifiability, capability onboarding under governance) to executive leadership on a cadence so the value is visible rather than invisible. Abandonment then requires someone to sign for the risk, rather than happening through quiet reallocation.

Function staffing pressure. The AEON function competes with the broader Enterprise IT function for senior talent. As Enterprise IT pressures intensify in the agentic era, the function may be raided for talent to address more visible problems. The concrete mechanism is to protect the function's senior roles explicitly in the Enterprise IT operating model, with the Chief Architect or equivalent holding the function's headcount as a protected allocation that cannot be raided for delivery surges without a documented decision at the same level that committed to the function.

Subdomain resistance to architectural authority. Subdomain teams accustomed to autonomy may resist the architectural contracts the function imposes. The resistance is sometimes legitimate (the contracts may not fit the subdomain's operational reality) and sometimes obstructive (the subdomain prefers not to be constrained). Distinguishing the two and addressing each appropriately is governance work that the function has to perform continuously. The concrete mechanism that makes the authority real is to tie AEON contract compliance to funding and project approval: new subdomain capabilities expose AEON-compliant adapters as a condition of funding, and capability qualification through the AEON lifecycle is a gate in the project approval process. Legitimate resistance is addressed by adapting the contract; obstructive resistance is addressed by the funding gate. The gate converts architectural authority from persuasion into a structural condition of the enterprise's investment process.

Policy authoring capacity. Translating enterprise policy into policy-as-code requires policy authoring capacity at scale. The capacity does not exist in most enterprises today; legal, compliance, and policy functions have not historically engaged with code-form policy expression. Building the capacity is itself a substantial program that runs in parallel with the AEON deployment.

Evidence storage cost. The evidence plane captures evidence at enterprise scope. The aggregate volume can be substantial, particularly with high-frequency subdomain activity. Storage cost, retention cost, and query cost have to be planned for; the architecture supports tiering and policy-driven retention, but the economic dimensions are real and have to be addressed in deployment planning.

Single point of architectural failure. The function exercises substantial architectural authority. If the function makes architectural errors, the errors propagate across the enterprise. The function has to operate with the discipline appropriate to the consequence of its decisions; this is institutional posture rather than technical architecture, but it is consequential.

17.3 Opportunities

Coherent enterprise governance. The most significant opportunity is the one the architecture is designed to deliver: coherent enterprise governance over agentic capability composition. Enterprises that achieve this position themselves to absorb the agentic era under institutional control rather than under vendor or accidental composition. The strategic value is substantial.

Cross-subdomain workflow capability. The integration plane makes cross-subdomain workflows architecturally first-class. Workflows that were previously impossible because they could not be composed across subdomain boundaries become possible. The capability is itself a competitive advantage in operations that require cross-subdomain reasoning.

Forensic and audit capability. The evidence plane makes cross-subdomain forensic and audit queries architecturally supported. Investigations and audits that previously required manual reconciliation across subdomain logs become standard queries against a coherent evidence picture. The reduction in audit overhead and the increase in audit reliability are both substantial.

Capability portfolio management. The capability composition function makes the enterprise's capability portfolio architecturally visible. Enterprise architecture functions that previously had to inventory capabilities through interviews and surveys can query the registry. Strategic capability decisions can be informed by accurate inventory rather than approximate guesses.

Institutional learning. The architecture-as-code practice produces durable institutional artifacts. The architecture, the policies, the workflows, the capabilities, the patterns are version-controlled, reviewable, and inheritable across team turnover. Institutional knowledge that previously existed in individuals' heads becomes architectural artifact that persists.

Strategic positioning in the agentic era. The enterprise that builds AEON well is positioned strategically for the agentic era. The capability becomes a competitive advantage in winning defense work that requires the governance posture AEON provides, in retaining workforce talent who want to work in well-architected environments, in absorbing acquired organizations under coherent governance, and in adapting to whatever the next wave brings.

Reference architecture leadership. Enterprises that build AEON well become reference architectures for the broader industry. The defense industrial base has historically led on accountable architecture for accountable institutions; AEON continues that tradition for the agentic era. The leadership has institutional value beyond the operational value of the architecture itself.

18. Conclusion

AEON is the enterprise control plane for the agentic era. It is the architectural layer that orchestrates capability composition across the enterprise's subdomains, owns the identity and authority fabric that flows across them, captures evidence at enterprise scope, decomposes policy into subdomain-specific enforcement, manages the capability lifecycle, and provides the meta-orchestration runtime the control plane requires. The architectural posture is institutional ownership: AEON is Enterprise IT-developed and sustained, with components supplied by vendors where appropriate and architectural authority belonging to the enterprise. The strategic claim is that the agentic era requires a control plane, that the control plane must be enterprise-owned to preserve institutional governance, and that Enterprise IT is the natural owner of the work.

The six planes the architecture specifies (identity, authority, evidence, integration, capability composition, and the meta-orchestration runtime that hosts them) compose into a control plane that subdomain runtimes consume through stable architectural contracts. The identity plane resolves principals across the federation and maintains delegation chain integrity across subdomain boundaries. The authority plane evaluates policy at enterprise altitude with seven decision types richer than binary allow-deny. The evidence plane aggregates canonical-schema evidence across subdomains into the enterprise audit picture and exposes worker-facing access as a worker-protection commitment. The integration plane orchestrates cross-subdomain workflows and the agent network. Capability composition governs the capability registry, lifecycle, patterns, versioning, and retirement. The meta-orchestration runtime is what Enterprise IT actually builds and operates: a coherent platform composed of vendor components under enterprise architectural authority.

The Enterprise IT operating model that builds and sustains AEON is a small, senior, architecturally focused function with the skill mix appropriate to control plane work. Identity engineering, policy engineering, evidence engineering, integration architecture, and capability composition are the disciplines the function practices. The function reports to the Chief Architect or equivalent, exercises architectural authority over subdomain contracts without authority over subdomain internals, operates the architecture as code, and runs the governance cadences that keep the architecture coherent over time. Building this function is itself a substantial program; the enterprise that commits to AEON commits to building and sustaining the function as the precondition for the work.

Classification environments are handled through the federated AEON instance pattern. Coordinated instances deploy across unclassified, CUI, NOFORN, and classified environments, with federation in identity, policy, evidence, and integration following defined patterns that respect classification directionality. The architecture is the same across instances; the federation between them follows policy controls that the enterprise establishes for its specific classification posture.

Composition with existing investments is the standard pattern rather than the exception. AEON overlays existing enterprise identity, policy, audit, integration, and platform investments through the federation broker, the policy compilation pipeline, the canonical evidence schemas, the adapter contract, and the capability registry. Existing investments are preserved and continue to deliver their existing value; AEON adds the cross-subdomain composition and enterprise governance that existing investments cannot provide. The migration sequence is selective and paced to the function's maturity.

The phased deployment establishes the architecture over a multi-year program. Phase zero establishes the function. Phase one builds the foundational identity and authority planes with pilot subdomain integration. Phase two adds the evidence and integration planes. Phase three formalizes capability composition and expands to additional subdomains and classification environments. Sustained evolution maintains and grows the architecture beyond initial deployment. Each phase has predictable failure modes that the program designs against; the countermeasures are institutional commitment, paced execution, and sustained operational discipline.

The opportunities are substantial. Coherent enterprise governance over agentic capability composition. Cross-subdomain workflow capability. Forensic and audit capability at enterprise scope. Capability portfolio visibility. Institutional learning encoded in architectural artifacts. Strategic positioning in the agentic era. Reference architecture leadership for the defense industrial base. The concerns are also substantial. Vendor capture by stealth. Institutional commitment erosion. Function staffing pressure. Subdomain resistance to architectural authority. Policy authoring capacity gaps. Evidence storage cost. Single point of architectural failure in the function itself. The architecture does not eliminate these concerns; it makes them visible enough to be addressed deliberately rather than discovered after they have damaged the deployment.

The defense industrial base has the institutional posture, the regulatory accountability, and the operational complexity that makes AEON not just appropriate but necessary. The agentic era is here; the question is whether the enterprise's control plane is deliberately architected or accidentally emerges from vendor decisions made without enterprise architectural intent. AEON is the framework for the deliberate version of the choice. The architecture is mature enough to commit to, the operating model is defined enough to staff, the phased deployment is sequenced enough to begin. The work is sustained, the commitment is institutional, the timeline is multi-year. The enterprise that begins this work in the present positions itself for the operational requirements of the next decade. The enterprise that defers the work finds itself absorbing the agentic era without the architectural means to govern it, which is not a position any defense enterprise should accept.

The next step is institutional commitment. Establish the function. Secure the senior architect. Build the initial team. Begin phase one. The architecture supports the work; the work supports the institution; the institution supports the strategic position the agentic era will reward.

Appendix A: The Subdomain Set

The nine peer subdomains AEON orchestrates are described below at the scope relevant to their composition under the control plane. Each subdomain has its own internal architecture, runtime, engineering function, and vendor relationships, which are outside the scope of this paper; what follows is the capability scope each subdomain exposes to the enterprise composition layer.

Governance and Strategy provides executive direction, strategic planning, portfolio management, investment governance, program management, compliance, risk, audit, and legal. It is the directing function of the enterprise, where the institution decides what it pursues and how it is held accountable. Compliance and risk are functions within this subdomain rather than peer subdomains, because they are tools the directing function uses rather than freestanding capabilities at the same altitude as the others.

AIDEX is the worker-facing AI digital experience substrate developed in the companion paper. It is the subdomain through which the next-generation knowledge worker engages AI-enabled capability under Human-Curated, AI-Enabled discipline. It consumes the AEON control plane more visibly than most subdomains because the worker's curation discipline is captured through the propose-and-confirm interactions the substrate mediates.

DevSecOps provides software delivery, CI/CD, security-integrated development lifecycle, release management, and pipeline governance. It is the subdomain through which the enterprise builds and delivers software, with security integrated into the delivery pipeline rather than bolted on after.

Cybersecurity provides security operations, threat management, information security, zero trust posture, identity security, and the agentic AI guardrails that constrain substrate behavior at the enterprise security altitude. It is the subdomain most directly concerned with the behavioral constraint layer the authority plane enforces, because the security implications of agentic action are its responsibility.

Digital Engineering provides model-based systems engineering, UAF and TOGAF practice, engineering models, digital thread, architecture repositories, and the engineering discipline that produces the systems the enterprise delivers. It is the subdomain where the enterprise's engineering knowledge is modeled and maintained.

Enterprise Platforms provides ERP, PLM, CRM, workflow platforms, the systems of record themselves, and the business process automation that runs on them. It is the subdomain that holds the enterprise’s authoritative business data and runs the business processes that operate on it.

Human Capital provides HR, talent management, learning and development, HCAE practice training, and workforce analytics. It is the subdomain through which the enterprise develops and sustains its workforce, including the practice training the HCAE discipline requires.

Insights and Intelligence provides data governance, data architecture, master data management, analytics, business intelligence, decision support, and embedded intelligence. It is the subdomain where the enterprise’s data is governed and where analytical and decision-support capability is produced. AIDEX consumes from this subdomain; this subdomain is broader than AIDEX because it includes capabilities that operate without direct worker interaction.

Infrastructure and Operations provides compute, network, storage, platform services, cloud and on-premise infrastructure, site reliability engineering, and the operational backbone everything else runs on, including the AEON runtime itself. It is the foundational subdomain on which all others, and AEON, operate.

Draft for discussion. Comments and challenges welcome.